

OCEAN SEQUESTRATION OF CO₂/N₂ MIXTURES

**A REPORT SUBMITTED TO THE DEPARTMENT OF
PETROLEUM ENGINEERING
OF STANFORD UNIVERSITY
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
DEGREE OR MASTER OF SCIENCE**

**By
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August, 2002**

I certify that I have read this report and that in my opinion it is fully adequate, in scope and in quality, as partial fulfillment of the degree of Master of Science in Petroleum Engineering.

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Abstract

CO₂ ocean sequestration provides a promising alternative for reducing greenhouse gas emissions provided an economic process is developed. The need to find a low cost process for long term CO₂ sequestration is the drive behind this study of the direct ocean release of CO₂/N₂ mixtures. Emitted flue gases are usually at most 15% CO₂ with the remainder consisting primarily of N₂. Processes that involve the direct release of pure CO₂ first require capture and separation of CO₂ from those flue gases. The direct release of low purity CO₂/N₂ mixture eliminates the costly step of separation. Furthermore, N₂ discourages the formation of hydrate, leading to greater dissolution rates at injection depths below 500 m.

This report discusses existing models of both pure CO₂ release and the P-GLAD model for low purity CO₂ injection developed by Saito *et al.* (2001). From the existing pure CO₂ direct release models a numeric model for direct mixture release is developed. The direct release of 15%, 50%, 85% and 100% from injection depths of 300, 500, 1000, and 1500 meters is simulated.

Finally the simulation results are used to compare the estimated cost of mixed gas direct release to that of the P-GLAD and pure CO₂ release. It is estimated that mixed gas direct release costs 118\$/ton CO₂. This is less than the estimate cost of liquid CO₂ release (126\$/ton CO₂), but double the estimated cost of the P-GLAD process (\$68/ton CO₂).

Acknowledgments

Firstly, thanks to Dr. Franklin Orr for his continuous support and guidance throughout this study. His confidence and trust in my ability enabled me to accomplish far more than I imagined.

Additional thanks go to Dr. Peter Brewer and the Monterey Bay Aquarium Research Institute for providing me with the unique opportunity to accompany them during field experiments. Nothing solidifies the equations and numbers more than observing them in action.

I am grateful to Dr. Khalid Aziz for his help understanding and simulating two-phase flow in pipes.

Thank you to all my friends who have listened to numerous exciting talks of gas bubbles and other such fascinations. I could not have made it through without their support.

I am incredibly indebted to Ray and Mark (and their kitchen table, at which the better portion of this was written). Thank you for doing so much without any obligation to do any of it at all.

Thank you, C. Ray, for helping me see the beauty outside my bubbles.

Finally, deep gratitude is due to my family and all those who have given their unconditional love and support throughout my life. Without my Mom and Dad I would never go the places I go, do the things I do, or learn the things I learn. To my sister, Sha, thank you for teaching me which standards are important.

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Chapter 1

1. Introduction

The energy usage of our growing worldwide population has resulted rising levels of greenhouse gas emissions. Over the past 150 years, atmospheric CO₂ concentrations have risen from 280 ppm to 370 ppm as a result of fossil fuel usage. It is estimated that CO₂ worldwide emissions in 1990 were greater than 20 Gt of CO₂ per year (DOE/ER-30194, 1994). It is possible that atmospheric CO₂ emissions will have to be limited if we are to continue to meet the energy demands of a growing population without contributing further to ecological decline. Current CO₂ sequestration research examines alternative storage locations for waste CO₂ and may provide cost-effective processes for CO₂ emissions reduction.

Geological formations provide opportunities for long term CO₂ storage. CO₂ can be dissolved into the salt water of deep aquifers. Additionally, oil and gas reservoirs provide capped porous rock formations for CO₂ storage accompanied by the possibility of enhanced oil recovery. Coal bed CO₂ injection provides similar benefits, with long term CO₂ storage accompanied by enhanced methane recovery.

In addition to geological CO₂ sequestration, current research is examining ways to take advantage of the ocean's large natural CO₂ storage capabilities. The DOE reports that the ocean occupies more than 70% of the Earth's surface, giving it huge potential for CO₂ storage (Golomb). Additionally, a major portion of atmospheric CO₂ eventually ends up in the ocean through natural processes and the quantity of CO₂ to be stored in the ocean is small relative to the quantity already there. It is estimated by the DOE that the ocean would show a 0.016% increase in CO₂ after disposal of all current annual carbon emissions (Golomb 1994). Proposed CO₂ ocean sequestration techniques can be divided into shallow water, midwater and deepwater releases. The CO₂ transportation and

dissolution processes of each differ due to oceanic conditions and resulting CO₂ phase behavior.

Whether geological or oceanic, nearly all current sequestration research explores storage of pure CO₂ gas. However, waste CO₂ gas is most frequently emitted in gas mixtures and the application of any such processes requires the separation of CO₂. Gas separation is a costly procedure, with cost estimates of one current chemical absorption method ranging from \$33 to \$43 per ton of CO₂ (Saito *et al.* 2001). Such costs make examining mixed gas sequestration, which explores the use of natural processes to eliminate the separation step, worthwhile. Such work is not only applicable to gas sequestration, but could also have widespread implications to all gas separation.

Recently, a team of Japanese researchers proposed the P-GLAD system for low purity CO₂ ocean disposal (Saito *et al.* 2001). The P-GLAD system consists of an inverse J pipeline and takes advantage of the gas lift effect to allow for shallow water injection of low purity gaseous CO₂ and deepwater release of dissolved CO₂. The system eliminates the costly steps of gas separation and CO₂ liquefaction yet provides the environmental advantage of deepwater disposal. The experimental and numerical work of Saito *et al.* ,(2001) demonstrates the economic advantage the P-GLAD system holds over direct liquid CO₂ release.

This report explores further mixed gas ocean release. It numerically examines direct ocean release of CO₂ and N₂ gas mixtures. It then compares these results to those found for the P-GLAD system. Chapter 2 provides background information on phase behavior and its role in ocean CO₂ sequestration. Chapter 3 and Chapter 4 provide a detailed discussion of the P-GLAD and open water bubble release models, respectively. They include discussions of the governing equations and assumptions of each model. Finally, Chapter 5 presents and discusses the simulation results and Chapter 6 highlights the conclusions of this study.

Chapter 2

2. Chemistry of Ocean Sequestration

Ocean sequestration involves complex systems that build on the basic concepts of phase behavior, multiphase flow, and mass transfer. This chapter begins with a discussion of hydrates and their impact on CO₂ sequestration. It then covers the basic phase behavior of CO₂, N₂, and CO₂/ N₂ mixtures. Finally, it shows the application of phase behavior in proposed CO₂ ocean sequestration technology.

2.1. Phase Behavior

Thermodynamics governs the properties of chemical components at varying pressures and temperatures. The temperatures and pressures of oceans, and the variations of those conditions with depth, makes phase behavior a key component to ocean sequestration. In the case of hydrocarbon ocean sequestration, low temperatures and high pressure induce the formation of hydrates. Hydrate formation must be accounted for in ocean sequestration models, as hydrates possess unique properties and alter mass transfer and bubble transport processes.

2.1.1 Hydrates

Hydrates are crystalline structures that form when hydrocarbons are in water at low temperatures and high pressures. Water molecules form a cage like structure which encases the hydrocarbon molecule. Much about hydrate behavior remains debated. For example, there is a great deal of uncertainty surrounding the exact formation, dissociation, and dissolution processes of hydrates. However, the importance of hydrate formation in CO₂ ocean sequestration has been recognized and has spurred a great deal of research.

In an early study on CO₂ dissolution behavior, Aya *et al.* (1992) report that the dissolution rate through a CO₂ hydrate coated bubble is lower than that through a bubble without hydrate. This idea is consistently supported throughout the literature. Figure 2-1 compares predictions from bubble models excluding hydrate formation and experimental data from Mori and Mochizuki (1998). The data in experiments where hydrate formation is not observed closely matches numerical predictions. However, the data collected when hydrate formation is observed shows slower bubble dissolution than predicted. Figure 2-2 compares the data from hydrate observed experiments to predictions using a modified modeling accounting for hydrate behavior. Brewer *et al.* (2000) finds even slower dissolution rates than those reported by Mori and Mochizuki (1998) during *in situ* deep sea bubble experiments.

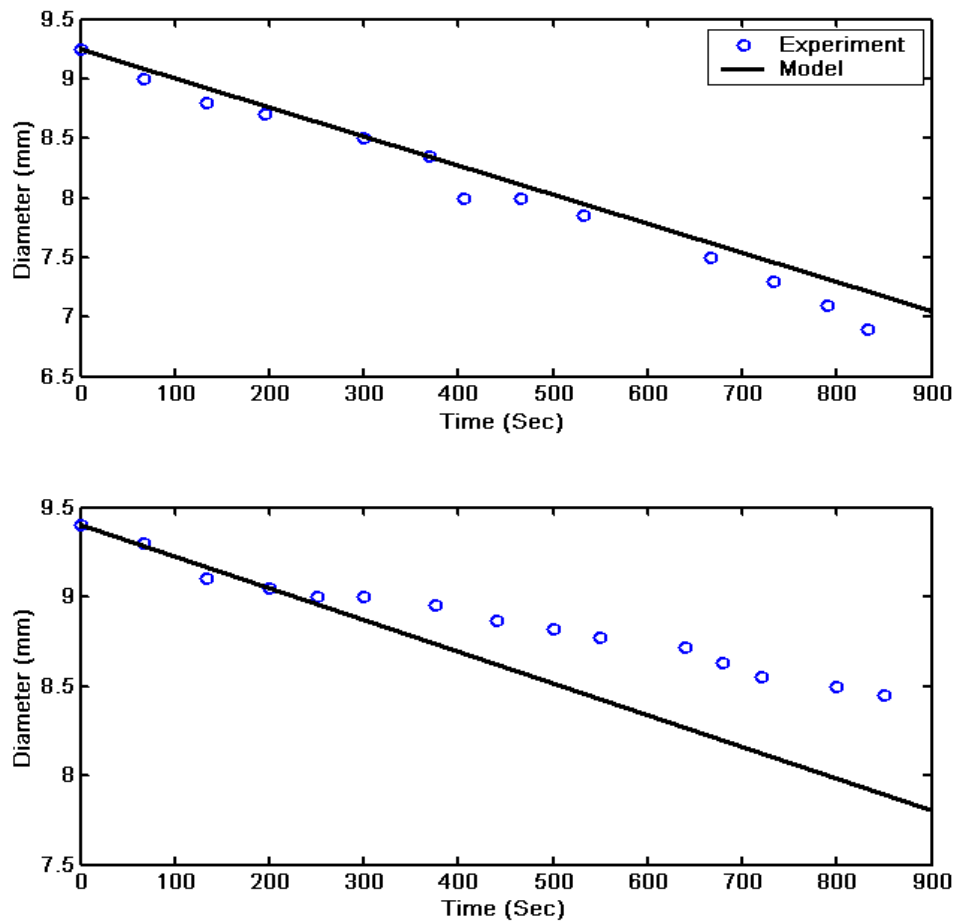


Figure 2-1: Experimental CO₂ bubble shrinkage, with and without hydrate formation (bottom), and predicted CO₂ bubble shrinkage (interpolated from Mori and Mochizuki, 1998)

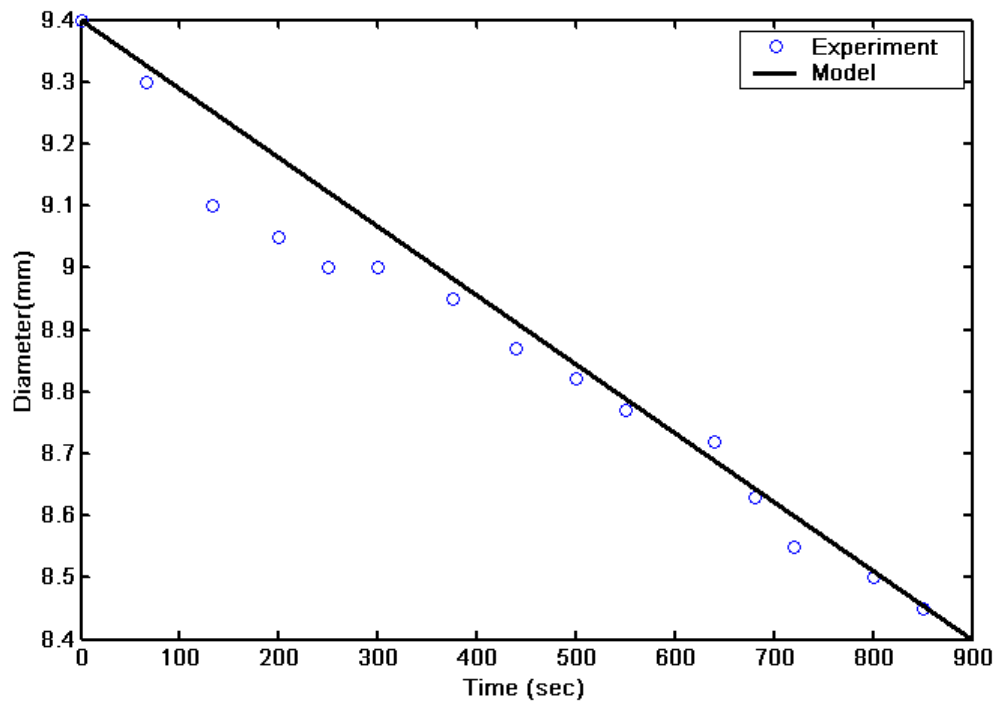


Figure 2-2: Experimental CO₂ hydrate coated bubble shrinkage and predicted CO₂ hydrate coated bubble shrinkage(interpolated from Mori and Mochizuki, 1998)

Brewer *et al.* (2000) points out that the formation of hydrates can have an important impact on the success of a sequestration project. Hydrate formation can benefit deepwater sequestration by increased sequestration time resulting from reduced CO₂ transfer rates. However, hydrate formation at shallower depths can result in ineffective sequestration if CO₂ is slowed enough that droplets reach the mixed zone without total dissolution. Failure to include and properly account for hydrate formation in a sequestration project can result in enormous economic costs.

2.1.2 Pure Carbon Dioxide Phase Behavior

Figure 2-3 graphs the gas-liquid transition line and the phase boundary for CO₂ hydrate formation for given depths and temperature. Phase behavior of CO₂ controls the density of CO₂ and thus the transportation process of injected CO₂ bubbles. Increasing depths lead to greater CO₂ densities. The rate of increase of CO₂ density with depth exceeds the rate of increase of seawater with depth and the density of liquid CO₂ becomes greater than

that of seawater around depths of 1800 m, as shown in Figure 2-4. Above this depth CO₂ droplets will rise and below this depth they sink.

Of equal importance is the hydrate formation zone, the range of depths and temperatures over which CO₂ and water form hydrate. It is important to note that CO₂ hydrate formation can be encountered over a wide range of depths and is not limited to solely deep waters.

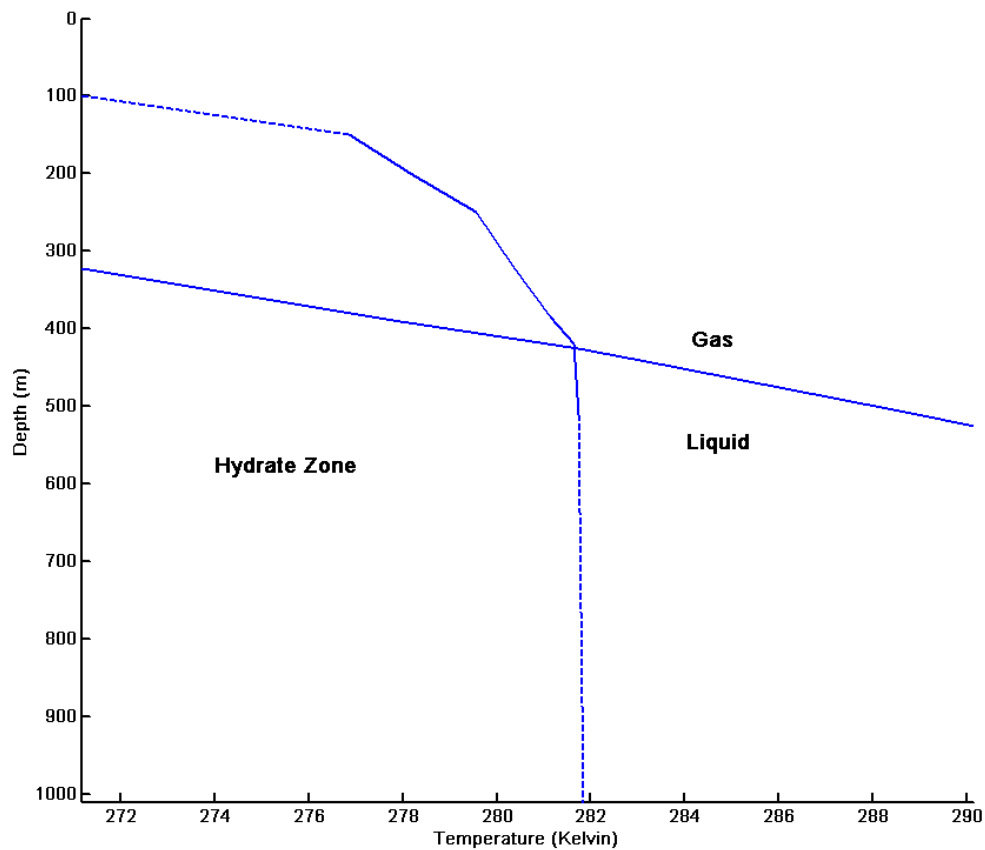


Figure 2-3: CO₂ phase behavior (adapted from Brewer *et al.*, 2000)

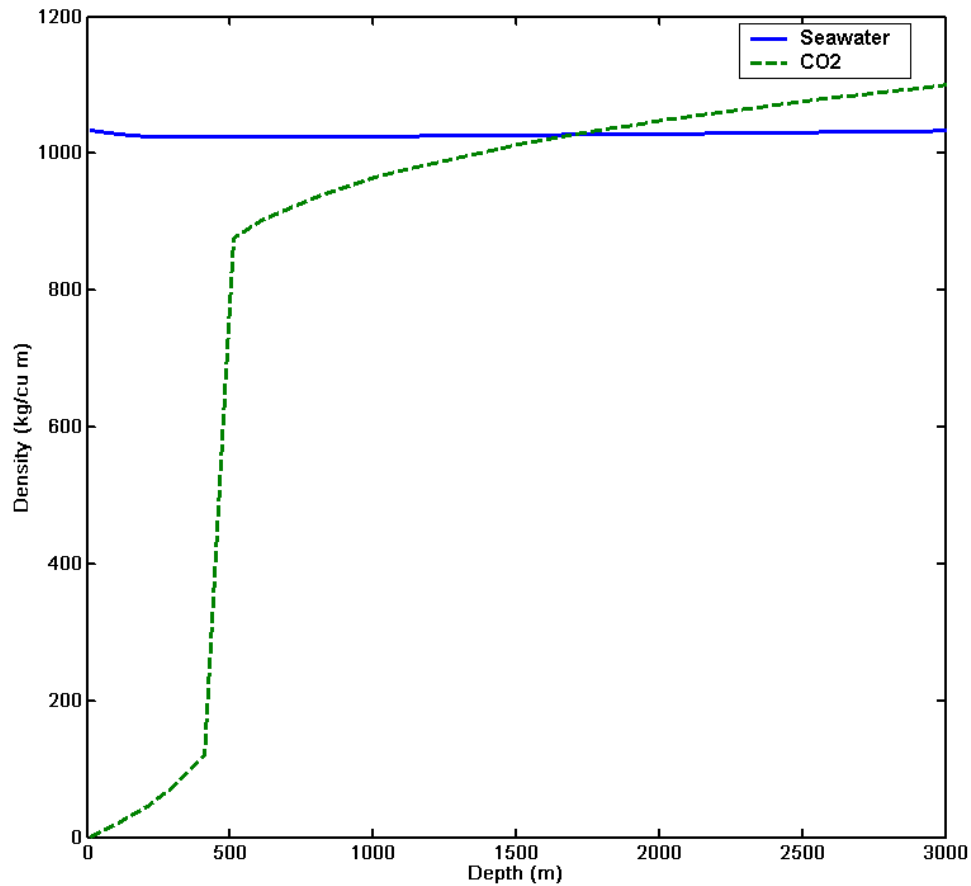


Figure 2-4: CO₂ and seawater density comparison (using temperature profile described in section 4.2)

2.1.3 Carbon Dioxide and Nitrogen Mixture Phase Behavior

The phase behavior of gaseous mixtures lies between the extremes of the pure components of the mixture. Nitrogen as a pure element is lighter and harder to compress than carbon dioxide. Furthermore, nitrogen hydrates are much harder to form than carbon dioxide hydrates. When nitrogen is added to carbon dioxide hydrate formation pressure at any given temperature increases. As a result, nitrogen and carbon dioxide mixtures have smaller hydrate zones than pure carbon dioxide.

Figure 2-5 compares predicted hydrate formation pressures for different nitrogen and carbon dioxide mixtures as predicted using the CSMHYD thermodynamic program

(Sloan, 1998). For a CO₂ content below about 50% mixture hydrate formation pressure drastically increases. As the mixture formation pressure increases for a given temperature, the hydrate formation zone decreases. Pure CO₂ will form hydrate at depths greater than 500 m at ocean pressures of approximately 50 atm and higher and ocean temperatures below about 279 K°. In contrast, a 50/50 mixture of CO₂/N₂ at 50 atm requires a temperature of 277 K° or less to form hydrate.

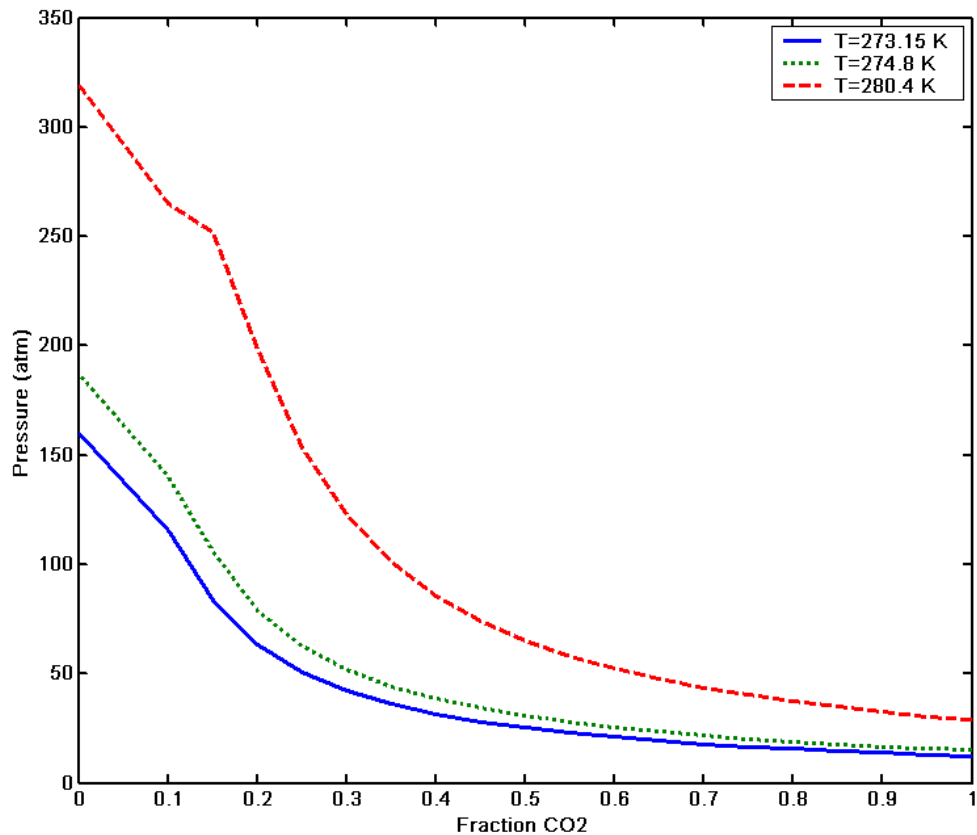


Figure 2-5: Variations in hydrate formation pressure of CO₂/N₂ mixtures in response to temperature and composition variations (Sloan)

A DOE report on CO₂ ocean sequestration reports a maximum CO₂ content of 15.4% in flue gases generated from coal, oil, and natural gas burnings. The hydrate formation zone of such mixtures is limited to extremely high depths.

2.2. Ocean Sequestration Processes

In shallow waters of 200-400 m, gaseous CO₂ is directly released through a pipe. The density of the CO₂ is less than that of the seawater and the gas bubbles rise. During the bubble ascent, CO₂ dissolves into the surrounding seawater. The upper mixed layer of the ocean extends approximately 100 m below the surface (Liro *et al.* 1992). To limit CO₂ release to the atmosphere through the ocean surface, any CO₂ released should be dissolved before reaching the upper mixed layer. Implementation of shallow water injection would presumably occur at ocean side power plants. Captured flue gases would be sent through a gas separation process. The extracted CO₂ could then be injected into the seawater through a gas pipeline. Shallow water injection offers the most economic capital, energy, and transportation costs, but its long term CO₂ storage capabilities are open to question.

Midwater release has been proposed for locations offshore in ocean depths of 400-1800 m. In midwater release processes, flue gases are captured, separated, and compressed. Liquefied CO₂ is then transported to the intended dumping site by ship and deposited at the desired depth through a transportation pipe. As discussed in the previous section, temperature and pressure conditions at depths greater than 400 m are such that CO₂ forms hydrates. Thus, the injected bubbles form an outer layer of hydrates and CO₂ dissolution occurs as the hydrate coated bubbles rise. Midwater release provides greater security for long term CO₂ storage. However, the compression and transportation of midwater release are costly. Saito *et al.* (2001) estimate the total cost of direct release of liquid CO₂ at depths near 1000m at \$118 per a ton of CO₂.

In addition to mid and shallow water release, storage of liquid CO₂ on the sea floor has been proposed. This is achieved by injection of liquid CO₂, after capture, separation and compression, at depths greater than 1800-2700 m depending on temperature profile. At such depths the density of liquid CO₂ is greater than that of seawater. Thus, injected CO₂ sinks to the sea floor. Deepwater injection, like midwater injection, is promising for long

term CO₂ storage. However, it is accompanied by even greater transportation and compressions costs than midwater release.

Considerable research and modeling of CO₂ ocean sequestration have recently been conducted and provide insight into the processes of CO₂ transportation and dissolution in the ocean. However, implementation of proposed ocean sequestration techniques has been limited, in part, by economic restraints. The need for an economic long term CO₂ sequestration procedure is the driving force behind the models discussed in the next two chapters. The P-GLAD system, proposed by Saito *et al.*(2000), limits compression and separation costs, but has higher capital and transportation costs than other methods. Mixed gas direct release limits separation costs, but may require greater compression costs. The aim of this study is, in part, to provide an economic comparison of the models as well as a preliminary assessment of the economics.

Chapter 3

3. P-GLAD Model

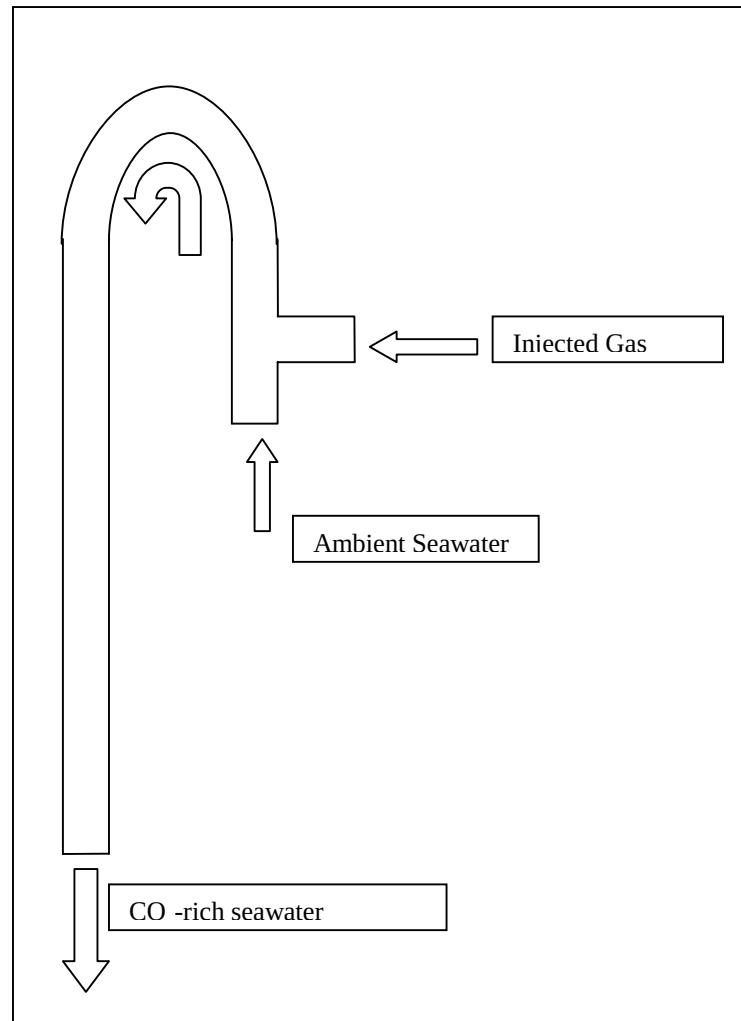


Figure 3-1: J-tube System (Saito *et al.* 2000)

Saito *et al.* (2000) propose a gas-lift J-tube system for the storage of low purity CO₂ mixtures. The system allows for shallow water injection of low purity CO₂, eliminating the cost of gas compression that accompanies deep-water injection. After injection, CO₂ is separated from the mixture and dissolves into surrounding seawater as the gas travels upward through the riser via gas-lift transport mechanisms. At the top of the riser, the indissoluble gases are emitted from the system upon reaching the top of the riser. The CO₂-rich seawater is then transported down the downcomer by the combined gas-lift and density differential effects. The Gas Lift Advanced Dissolution system proposed by Saito *et al.* (2000) provides a comparatively low cost method for CO₂ sequestration by eliminating the necessity of pre-injection separation and gas compression. Additionally, the system provides the long term security of deepwater deposition and CO₂ storage.

3.1 Governing Equations

3.1.1 Conservation Equations

The GLAD system is a gas-liquid two-phase flow model including gas dissolution. The governing equations are laws of conservation of mass and momentum and the gas equation. The gas flow is assumed to be bubbly flow of uniform size. The assumption of constant temperature eliminates the necessity of the energy equation. All equations found in this section are discussed in Saito *et al.* (2000).

The density and mass of the liquid and gas mixture (ρ_M and $\rho_M v_M$) are defined as:

$$\rho_M = \alpha \rho_G + (1 - \alpha) \rho_L \quad \text{and} \quad \rho_M v_M = \alpha \rho_G v_G + (1 - \alpha) \rho_L v_L \quad (3-1)$$

where v is the phase velocity, α is the gas void fraction, and the subscripts G and L refer to the gas and liquid phases respectively.

A mass and force balance yield the conservation of mass and conservation of momentum equations for the mixture:

$$\frac{\partial \rho_M}{\partial t} + \frac{\partial(\rho_M v_M)}{\partial z} = q_G \quad (3-2)$$

and

$$\frac{\partial(\rho_M v_M)}{\partial t} + \frac{\partial(\alpha \rho_G v_G^2)}{\partial z} + \frac{\partial[1-\alpha]\rho_L v_L}{\partial z} = -\frac{\partial P}{\partial z} - \rho_M g - \frac{4\tau_w}{D_p} \quad (3-3)$$

where P is the static pressure [Pa], q_g is the gas injection rate per unit volume [$\text{kg}/(\text{m}^3\text{s})$], τ_w is the friction stress at the wall [Pa], and D_p is the diameter of the dissolution pipe [m]. A mass balance on gas must be included to account for gas dissolution:

$$\frac{\partial(\alpha \rho_G)}{\partial t} + \frac{\partial(\alpha \rho_G v_G)}{\partial z} = q_G - \Gamma_G \quad (3-4)$$

where the gas dissolution rate Γ_g is given by

$$\Gamma_G = 4\pi r_b^2 n_b k_L M_g (C_s - C_\infty) \quad (3-5)$$

and r_b is the bubble radius [m], n_b is the number bubbles, k_L is the mass-transfer coefficient, M_g is the molar mass of gas [mol], C_s is the saturated molar concentration [mol/m^3], and C_∞ is the molar concentration in the liquid [mol/m^3].

Assuming a conservative system of bubbles, a mass balance on the number of bubbles is performed:

$$\frac{\partial n_b}{\partial t} + \frac{\partial(n_b v_G)}{\partial z} = \frac{q_G}{m_{G0}} \quad (3-6)$$

where m_{G0} is the initial mass of injected bubbles[kg].

3.1.2 Bubble Transportation: Drift Flux Equations

The drift flux model describes the multiphase flow within the system pipes. The drift flux model and its accompanying equations found in this section are discussed by Chen (2001).

The superficial velocities of each phase (J_L and J_G) are defined by:

$$J_L = \frac{Q_L}{A} \text{ and } J_G = \frac{Q_G}{A} \quad (3-7)$$

where Q_L and Q_G are the volumetric flow rates of the liquid and gas phases, respectively, and A is the pipe cross sectional area.

The total mixture velocity given by

$$J_M = J_L + J_G. \quad (3-8)$$

In multiphase flow, differing phase properties, like density and velocity, result in unique phase velocities. The different velocities lead to the “slip” and “hold-up” of the phases relative to one another. Consequently, the *in situ* void fractions are not equal to the initial injection fractions (Chen 2001). The *in situ* gas volume fraction α is given by:

$$\alpha = \frac{A_G}{A} \quad (3-9)$$

where A_G is the area occupied by gas.

The average *in situ* velocities of each phase are calculated from the *in situ* volume fraction:

$$u_L = \frac{J_L}{(1-\alpha)} \quad (3-10)$$

and

$$u_G = \frac{J_G}{\alpha}. \quad (3-11)$$

Finally, Saito *et al.* (2001) use the following correlation between *in situ* gas velocity and total mixture velocity:

$$u_G = B_0 J + U_G \quad (3-12)$$

where

$$B_0 = 1.2 - 0.2\sqrt{\rho_G/\rho_L} \quad (3-13)$$

and

$$U_G = 1.41[\sigma_L g(\rho_L - \rho_G)/\rho_L^2] \quad (3-14)$$

and σ_L is the interfacial tension.

3.2 Calculating Mass Transfer Coefficient

The mass transfer coefficient is a key factor in the numerical J-tube model. Saito *et al.* (2000) conducted small scale bubble visualization experiments to determine a mass transfer coefficient. The experiments utilized video equipment to observe bubble interfacial area and volume of bubbles over time. Additionally, average bubble rise velocity was measured. Calculating the mass transfer coefficient, k_L , from the data collected starts with:

$$\frac{dC_B(t)}{dt} = -k_L(t)A_T(t)[C_s(t) - C_\infty(t)] \quad (3-15)$$

where $C_B(t)$ is mol of the bubbles, $A_T(t)$ is the total interfacial area of the bubbles, $C_s(t)$ is the CO_2 molar concentration of the surrounding liquid (saturation solubility of CO_2 in water at pressure $P(t)$), and C_∞ is the molar concentration of CO_2 in the liquid phase (Saito *et al.*, 2000).

$C_B(t)$ is given by

$$C_B(t) = [P_B(t)V_T(t)]/\beta RT \quad (3-16)$$

where β is the compressibility factor and R is the gas constant (Saito *et al.*, 2000).

The average pressure inside the bubble, $P_B(t)$ is determined from

$$P_B(t) = P(t) + 4\sigma_L/D_B(t) \quad (3-17)$$

$D_B(t)$ is the average bubble diameter and $P(t)$ is the measured static pressure (Saito *et al.*, 2000).

Finally, $C_\infty(t)$ is given by

$$C_\infty(t) = C_o + [C_B(t_o) - C_B(t)] \quad (3-18)$$

$$A_T(t) = n(t)A_B(t) \quad (3-19)$$

$$V_T(t) = n(t)V_B(t) \quad (3-20)$$

where C_o is the initial CO_2 concentration of water, $A_B(t)$ is the measured average interfacial area of the bubbles, $n(t)$ is the measured number of bubbles, and $V_B(t)$ is the measured volume of bubbles (Saito *et al.*, 2000). Through the experiments Saito *et al.*

calculate a constant mass transfer coefficient of 2.0×10^{-4} m/s, which is used throughout their numerical computations they use the

3.3 Simplified Model

The model used by Saito *et al.* (2001) provides accurate, detailed results, including unsteady flow but at the significant cost of complexity. This study examines a simpler model that begins by separating the J-tube into two sections, the upriser and the downriser. Flow in the upriser is calculated using concepts of multiphase flow in a pipe. Flow in the downriser is a simpler single phase flow problem controlled by the conservation of mass.

3.3.1 The Upriser: Multiphase Flow in a Pipe

Modeling flow in the upriser begins by considering the pressure gradient during multiphase flow in a vertical pipe:

$$\Delta P = \frac{M_L + M_G}{Q_L + Q_G} \left(\frac{g}{g_c} \right) \Delta z + \Delta P_f \quad (3-21)$$

where M is the mass flow rate, Q is the volumetric flow rate, and subscripts L and G represent liquid and gas respectively (Govier and Aziz, 1972). ΔP_f is the pressure drop due to irreversibilities over Δz . It includes the effects of friction and holdup, amongst other things, within the pipe.

ΔP_f is estimated by correlations developed through data collection. There are numerous correlations that cover a range of pipe sizes, flow regimes, and fluid pairs. They fall into two major categories, those that estimate fluid conditions as functions of distance within the pipe and those that do not. To maintain simplicity, this model uses the correlations of Govier *et al.* (1972) from the later category.

The correlations of Govier *et al.* (1972) were developed from data collected on an air-water system and cover a wide range of pipe diameters and flow regimes. The correlation

plots the two-phase friction flow factor versus a reduced liquid Reynolds number, DV_{SL} , and $D^{3.5}V_{SG}$ (V_{SL} and V_{SG} are the superficial liquid and gas velocities, respectively and D is the pipe diameter). ΔP_f is calculated from the relationship

$$\frac{\Delta P_f}{\Delta z} = \frac{2\rho_L V_{SL}^2}{g_c D} f_{tpL} \quad (3-22)$$

where ρ_L is the liquid density and f_{tpL} is the two phase friction factor given by the Govier correlation. The data assumes constant pressure and temperature and is therefore best suited for point specific calculations (Govier and Aziz, 1972). Therefore, in this model the upriser is divided into small 5 meter sections. The fluid densities, used to calculate volumetric flow rates and superficial velocities, are calculated at each section using the Peng Robinson equation of state. The friction factor is then determined for each section and used to calculate a section specific ΔP_f .

3.3.2 The Downriser: Single Phase Flow in a Pipe

It is assumed that all gases either dissolve into the seawater or are emitted through the top of the J-tube. Thus, flow in the downriser is single phase flow of CO_2 saturated seawater. The pressure gradient for such flow is given by:

$$\Delta P = \frac{g}{g_c} \Delta z \rho + \frac{\rho}{2g_c} \alpha (V_2^2 - V_1^2) + \frac{32\mu V L_E}{g_c D^2} \quad (3-23)$$

where μ is the fluid viscosity, V is the fluid velocity, and L_E is the length of the pipe. For the J-tube model the kinetic energy term, the second term of equation 4-23, is negligible and can be omitted. To further simplify the model, the fluid properties, viscosity and density, are assumed to be constant throughout the length of the downriser.

3.3.3 *Boundary and Initial Conditions*

The initial conditions of the J-tube model differ from typical well applications in that the flow rate of water is not specified. The rate of injection of gas is controlled but the amount of seawater is not pumped into the system at a controlled rate. Instead, the flow of water is determined by the boundary pressures at the pipe openings and the amount of water the injected gas is capable of lifting. The model upriser is 200 meters long with a boundary at 300 meters below the surface. The downriser is 900 meters long with a boundary 1000 meters below the surface.

3.3.4 *Solving the Model*

The pressure gradient equations and the initial conditions are used to solve for the J-tube's pumping ability, captured by the fluid velocity variable. A guess and check procedure is used to find the fluid velocity that correctly solves the pressure gradient equations, setting the pressures at the bottom of the upriser and the bottom of the downriser to correct hydrostatic pressures.

The 200 meter upriser is divided into 5 meter sections. Fluid properties are determined for each section and used along with the gas injection rate and the guessed flow rate of seawater to solve the pressure gradient equation in the upriser. Conservation of mass requires that all the water flowing from the upriser flows into the downriser. Thus, the velocity of the water at the top of the downriser must equal the velocity at the top of the upriser. This velocity is used to solve the pressure gradient in the downriser, which is assumed to be one long pipe with constant fluid properties. The predicted pressure at the bottom of the downriser is then checked against the actual hydrostatic pressure. If the two do not match, a new water flow rate is guessed and the procedure is repeated.

Chapter 4

4. Single Bubble Release

The single bubble release model examines the transportation and dissolution processes that occur with the release of a single bubble of gas and various depths. The model predicts the velocity, bubble diameter and CO₂ concentration over time. Several studies have numerically modeled the release of pure CO₂ bubble into the ocean. Mori and Muchizuki (1998) examined the behavior of pure CO₂ bubbles in seawater with the intention of better understanding the effect of hydrate formation on CO₂ dissolution rates. Hirai *et al.* (1999) also provide a model of pure CO₂ direct ocean release in their study on the environmental impacts of CO₂ ocean sequestration. The past models of pure CO₂ release provide a basis upon which to build a model of mixed gas bubble release.

Bubble release models are composed of two processes, gas dissolution and bubble transport. To accurately model bubble release the governing equations of both processes and the procedure to simultaneously solving those equations must be understood. Also essential to solving the model is correct evaluation of its parameters and properties.

4.1 Governing Equations

4.1.1 Bubble Transportation

Bubble rise velocity is dependent on numerous factors, including bubble shape, density, and size as well as the properties of the surroundings through which it travels. In the case of ocean gas injection it is reasonable to assume spherical shapes with negligible deformation. Yabe *et al.* (1999) examine the validity of this assumption by examining the Weber number, the dimensionless number directly related to bubble deformation. They found a maximum Weber number in their experiments, 0.17, below the Weber number, 0.5, indicative of bubble deformation. Additionally, the numeric models of Mori and Mochizuki (1998) for rigid spheres more closely match experimental data than those for fluid spheres.

The velocity of the bubble can be found by performing a force balance on the assumed spherical bubble:

$$\frac{1}{6}\pi D_b^3 \left(\rho_g + \frac{1}{2}\rho_w \right) \frac{dU}{dt} = \frac{1}{6}\pi D_b^3 (\rho_w - \rho_g)g - \frac{1}{8}\pi D_b^2 \rho_w U^2 C_D \quad (4-1)$$

where U is the free rise velocity and C_D is the drag coefficient for a sphere at steady velocity U .

The drag coefficient, C_D , is calculated from the correlation for a rigid sphere (Mori and Mochizuki, 1998):

$$C_D = \frac{24}{\text{Re}} \left(1 + 0.173 \text{Re}^{0.657} \right) + \frac{0.413}{1 + 16300 \text{Re}^{-1.09}} \quad (4-3)$$

where $\text{Re} \equiv \frac{UD_B}{\eta_w}$ and η_w is the seawater viscosity.

Solving for U ,

$$\frac{dU}{dt} + \frac{3}{4}\xi \frac{C_D}{D_b} U^2 - \xi(1-\gamma)g = 0 \quad (4-2)$$

where $\gamma \equiv \frac{\rho_g}{\rho_w}$ and $\xi \equiv \frac{\rho_w}{\left(\rho_g + \frac{1}{2}\rho_w \right)}$.

The interdependence of Re and U makes an iterative procedure necessary to solve for U . This model solves a discretized form of equation 4-1, using an approximation of the time derivative, to numerically solve for rise velocity. The result is the following quadratic equation:

$$\left(\frac{3}{4} \xi \frac{C_D}{D_b^{n+1}} \right) U^{n+1^2} + U^{n+1} - [\Delta t \xi (1 - \gamma) - U^n] = 0 \quad (4-4).$$

This equation is directly solved for the velocity within a time step. This is accomplished by first guessing a rise velocity and calculating the corresponding Reynolds number. The Reynolds number is in turn used to determine the corresponding drag coefficient. A rise velocity is then calculated using equation 4-4 and the guessed drag coefficient. The calculated velocity's corresponding Reynolds number is determined and checked against the guessed velocity's Reynolds number. If the two do not match, the calculated velocity becomes the new guessed velocity and the procedure is repeated.

4.1.2 Mass Transfer

Mori and Mochizuki (1998) sought to understand the mass transfer of CO₂ from the bubble to the surrounding seawater by first studying just the shrinking of a stationary CO₂ bubble over time. The model is governed by the conservation of mass of CO₂ yielding:

$$-\frac{d}{dt} \left(\frac{1}{6} \pi D_b^3 \rho_g \right) = \pi D_b^2 \alpha_D \rho_w x_{gs} \quad (4-5)$$

where D_b is the bubble diameter, α_D is the mass transfer coefficient, and x_{gs} is the CO₂ solubility in seawater.

The right hand side of equation 4-5 multiplies the bubble volume by bubble density resulting in the change of mass of the bubble over time. It is important to recognize that bubble diameter is not the only parameter that changes with time. As the bubble rises it will face changing thermodynamic conditions that lead to both changing bubble and seawater densities. In the case of the mixed gas bubble, different components will have different solubility in seawater which results in a varying bubble composition. This

further contributes to the change in bubble density over time and leads to a varying mass transfer coefficient, as well.

Mori and Mochizuki (1998) solve equation 4-1 for the rate of change of diameter with time. With a pure gas bubble model it is not necessary to know the rate of change of mass over time. However, in the case of a mixed gas bubble the change in composition over time must be calculated. Because volume is not conserved, diameter change is not directly representative of the rate of mass loss but is rather a result of the combined thermodynamic changes and gas dissolution. Therefore, determining the change in bubble concentration requires the change in mass.

This model calculates the change in mass in a given time step. It is assumed that the solubility of N₂ in seawater is negligible, and thus the entire quantity of mass lost is CO₂. This assumption is reasonable as the solubility of nitrogen in seawater is magnitudes of order below that of carbon dioxide. The entire quantity of mass lost is easily converted to a molar quantity, from which the change in composition is calculated. Using the predicted density and bubble composition at the next time step the bubble diameter is predicted.

4.2 Determining Seawater Properties

Oceanic temperature and pressure are a function of depth. The seawater temperature, in degrees Kelvin, is calculated using a temperature and depth correlation referenced by Ohmura and Mori (1998):

$$T = \frac{3946}{z + 112} + 273.6 \quad (4-6).$$

Seawater pressure and density are determined using Morgan's SEAWATER computational routines written for MATLAB (1994). All calculations are determined at a latitude of 36.75, approximately that of Monterrey Bay, CA. Figure 4-1 displays the

ocean temperature and pressure profiles. As the figure demonstrates, pressure variation with depth is much greater than temperature variation. Therefore, in many seawater calculations, particularly those that involve only small depth variations, temperature is assumed constant.

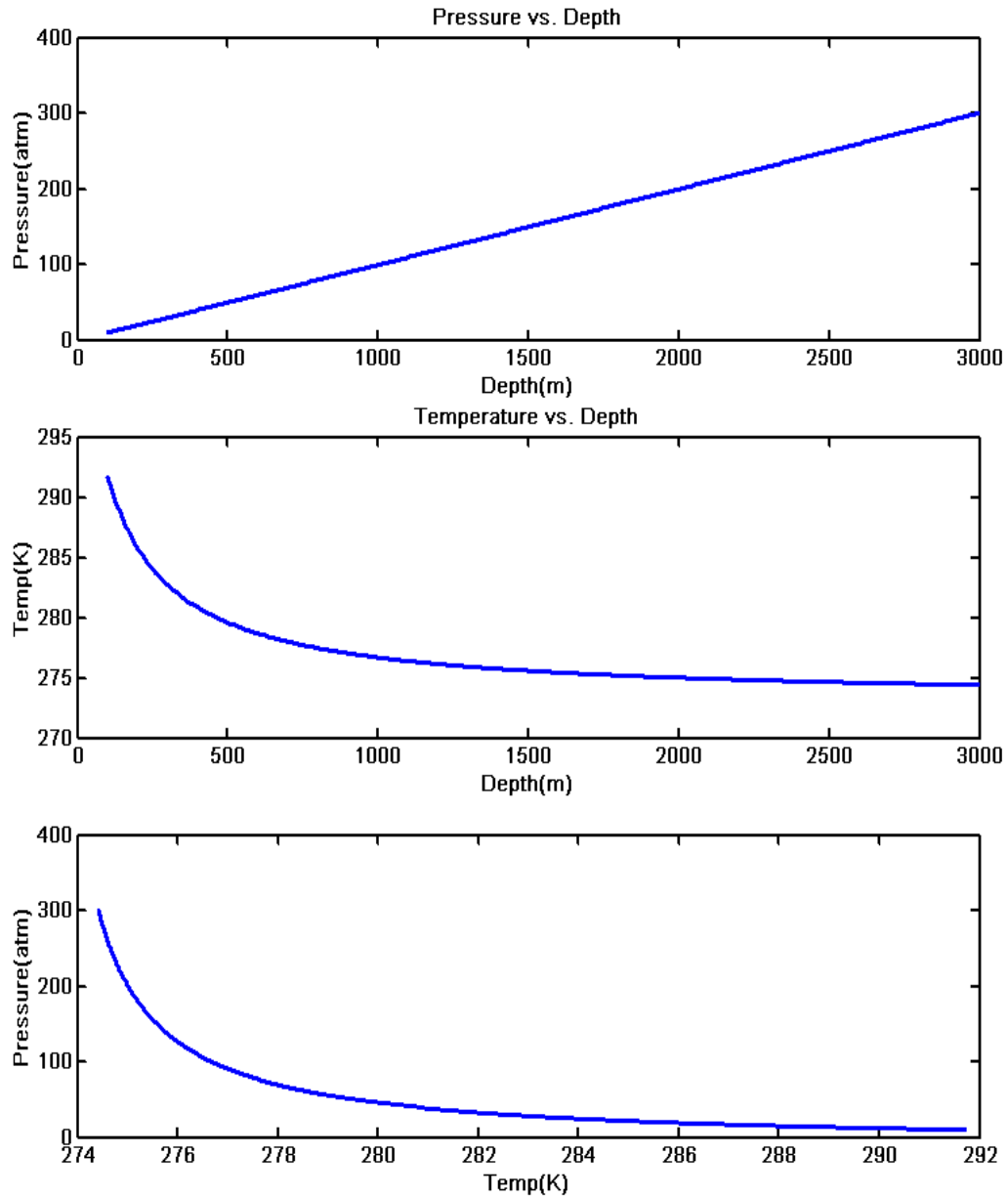


Figure 4-1: Oceanic temperature and pressure profiles (SEAWATER 1994)

4.3 Determining Bubble Properties

In this model the presence of nitrogen eliminates the formation of hydrate, and all bubble properties are simply gas mixture properties. The dissolution of carbon dioxide into the seawater implies a varying bubble composition with time. Furthermore, the change in bubble depth implies changing pressure and temperatures. Therefore, bubble properties must be recalculated at each time step at the current composition, temperature, and pressure.

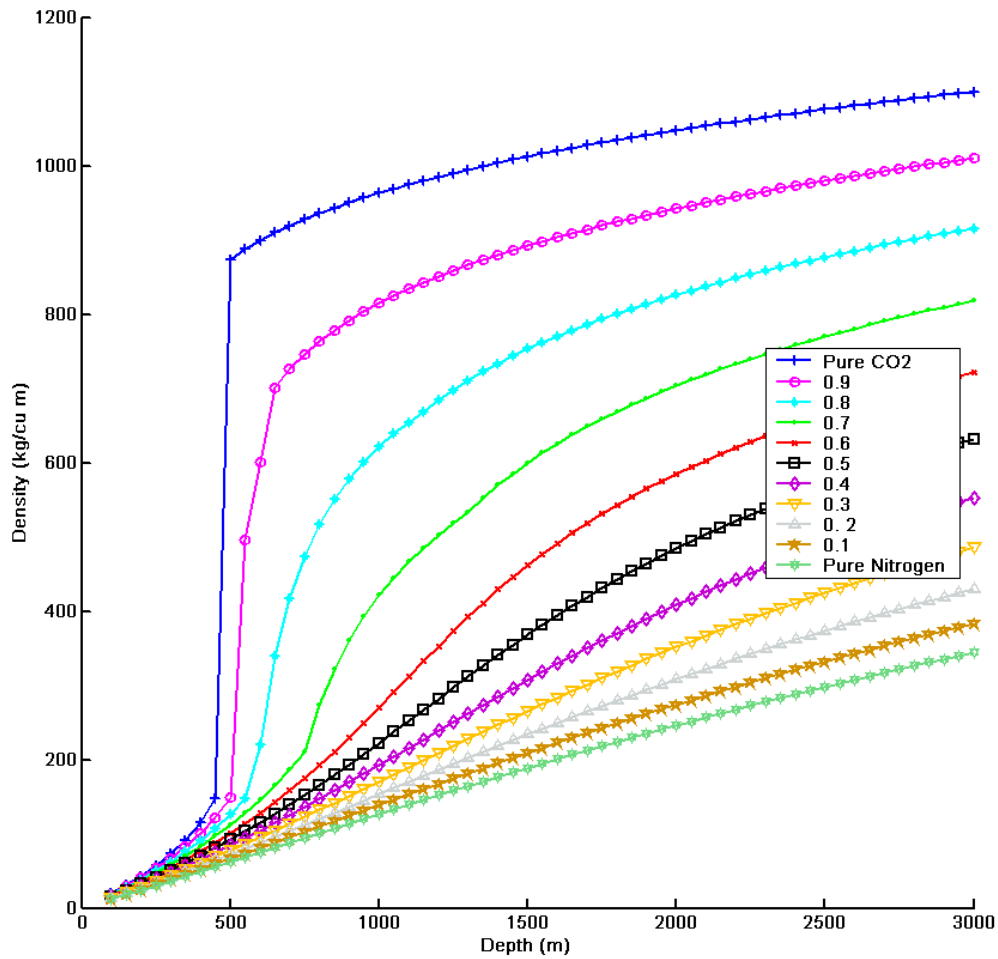


Figure 4-2: Variation in mixture density resulting from CO₂/N₂ concentration variation for the temperature profile given in Figure 4-1

4.3.1 Bubble Density

The Peng Robinson equation of state is used to determine the bubble phase behavior based on the ocean temperature and pressures at the current bubble depth. Figure 4-2 displays the density of various composition bubbles. The figure shows decreasing densities with decreasing concentration of carbon dioxide. This indicates that increasing nitrogen concentration will increase the specific volume of a mixture, implying rising compressions costs with rising nitrogen concentrations.

4.3.2 Carbon Dioxide Solubility in Seawater

Determining the mass solubility of carbon dioxide in seawater, x_{gs} in this model, is simplified by the lack of hydrate formation. Though no exact formula is available, much work experimental work has been done studying the change in carbon dioxide solubility for changing seawater pressures and temperature. This model interpolates the data used by Mori and Mochizuki (1998) in their pure CO₂ release simulations. The solubility profile with depth (using the temperature profile of figure 4-1) is plotted in Figure 4-3.

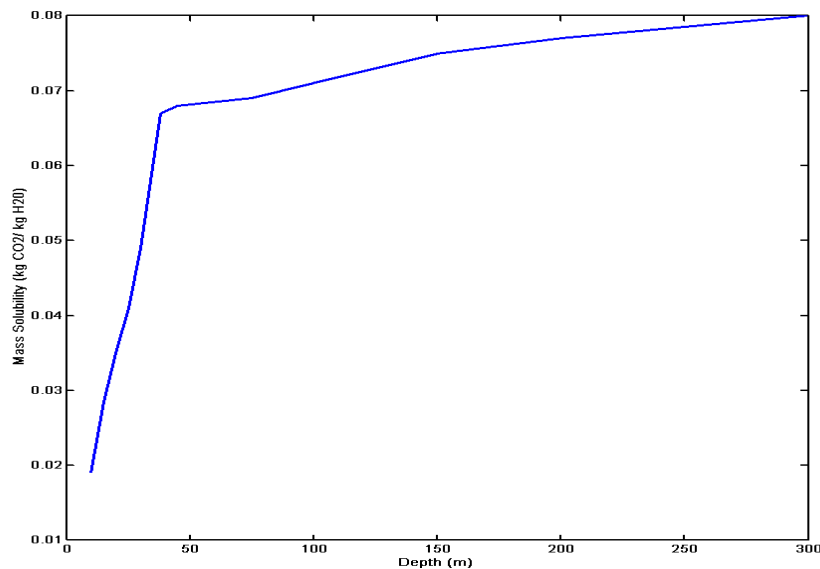


Figure 4-3: CO₂ solubility in seawater (interpolated from Mori and Mochizuki, 1998)

4.3.3 Mass Transfer Coefficient

4.3.3.1 REPORTED MASS TRANSFER COEFFICIENTS

A good deal of discrepancy surrounds the rate of mass dissolution of CO₂ into seawater. Furthermore, methods of measuring and reporting the CO₂ mass transfer coefficient are inconsistent. Using reported mass transfer coefficients requires attention to the details of their source.

Mori and Mochizuki (1998) estimate the value of the mass transfer coefficient using empirical correlations and a modified Wilke-Chang equation to determine the coefficient of diffusion of CO₂ in water. The correlations define the mass transfer rate as a function of bubble diameter and rise velocity and seawater viscosity and density. The result is a mass transfer rate that varies over time and is strongly dependent on bubble velocity. The study examines diameter change over time, and thus the coefficient is reported in units of m/s.

Experimental work suggests a constant mass transfer coefficient. The constant coefficient of 2×10^{-4} m/s used in the P-GLAD simulation by Saito *et al.* (2000) is one of the higher determined mass transfer coefficients and does not include hydrate formation. Hirai *et al.* (1996) report a mass transfer rate of 1.25×10^{-5} kg m⁻² s⁻¹ for non-hydrate coated bubbles.

A greater amount of work deals with the mass transfer process of hydrate coated bubbles. Brewer *et al.* (2000) provide a mass transfer coefficient on the lower end of the scale, based on sea experiments, of 3.4×10^{-6} mol m⁻² s⁻¹ for hydrate coated bubbles. Two separate reports, Fujioka *et al.* (1994) and Aya *et al.* (1992), report bubble diameter reduction rates of 5.0×10^{-7} and 3.9×10^{-9} m/s, respectively. Tabe *et al.* (1999) report a constant mass transfer rate of CO₂ through hydrate of 1.15×10^{-4} kg m⁻² s⁻¹. Hirai *et al.* (1996) report varying mass transfer rates through hydrate depending on temperature, pressure, and rise velocity. Their experimentally measured rates range from

approximately 3×10^{-4} to $1.5 \times 10^{-3} \text{ kg m}^{-2} \text{ s}^{-1}$, but they offer no correlations for calculating the rates.

A further complication in this model is the varying compositions of the bubbles. Saito *et al.* make no change in their mass transfer coefficient to adjust for impurities in the injected gas. However, as N_2 increases the mass transfer rate should decrease. In this study it is assumed that the mass transfer coefficient for the mixed gas bubble is linearly proportional to the concentration of CO_2 in the bubble.

Table 4-1 summarizes different experimentally calculated mass transfer coefficients for bubbles with and without hydrate, respectively. If possible with the reported information, the rates have been converted into units of $\text{mol m}^{-2} \text{ s}^{-1}$ for an 85% pure carbon dioxide bubble at a depth of 1500 m.

Table 4-1: Reported CO_2 Mass Transfer Coefficients

Source	Molar Transfer Coefficients [$\text{mol m}^{-2} \text{ s}^{-1}$]	Hydrate
Brewer <i>et al.</i> (2000)	3.4×10^{-6}	yes
Aya <i>et al.</i> (1992)	7.6×10^{-5}	yes
Tabe <i>et al.</i> (1999)	3.4×10^{-3}	yes
Hirai <i>et al.</i> (1996)	1.1×10^{-7} to 4.7×10^{-7}	yes
Fujioka <i>et al.</i> (1994)	2.4×10^{-5}	yes
Saito <i>et al.</i> (2000)	2.0×10^{-4} (m/s)	no
Hirai <i>et al.</i> (1996)	2.8×10^{-4}	no

4.3.3.2 THIS MODEL

This report compares two different methods of calculating the mass transfer coefficient. Method A follows the procedure used by Mori and Mochizuki (1998) and uses empirical correlations to calculate the mass transfer coefficient. Method B uses the mass transfer rate reported by Hirai *et al.* (1996) for non-hydrated coated bubbles. Both models incorporate a modified mass transfer coefficient for pure CO₂ over depths of 500 m to account for hydrate formation. This is done using the diameter reduction rate reported by Fujioka *et al.* (1994) and equation 4-5 to calculate the mass transfer coefficient given time specific bubble properties.

Method A:

Mori and Mochizuki estimate the value of α_D using correlations for the dimensionless Sherwood number, Sh , which is defined as $\alpha_D D_b / D_{gw}$. The Sherwood number is determined from correlations for a rigid sphere:

$$Sh = 1 + 0.425 Re^{0.55} Sc^{1/3} \quad (Re \geq 2 \times 10^3) \quad (4-6)$$

$$Sh = 1 + 0.724 Re^{0.48} Sc^{1/3} \quad (100 \leq Re \leq 2 \times 10^3) \quad (4-7)$$

where

$$Re \equiv \frac{U d \rho_w}{\eta_w} \quad (4-8)$$

and

$$Sc \equiv \frac{\eta_w}{\rho_w D_{gw}} \quad (4-9).$$

D_{gw} , the mass diffusion coefficient of CO₂ in water, is calculated using the Wilke-Chang equation modified as suggested by Hayduk and Laudie (1974):

$$D_{gw} = \frac{7.4(10^{-8})(2.26 * M_2)^{0.5} T}{\mu_2 V_1^{0.6}} \quad (4-10)$$

where M_2 is the molecular weight of the solvent (water), T is the temperature in K, μ_2 is the viscosity of the solvent, and V_1 is the molecular volume of the solute (CO_2).

Method B:

The mass transfer rate reported by Hirai *et al.* (1996) is converted to a molecular transfer rate using the molecular weight of CO_2 :

$$\alpha_D = 1.25 \times 10^{-4} \pi D_b^2 X_{CO_2} \left(\frac{1000}{44.01} \right) \left[\frac{mol CO_2}{s} \right] \quad (4-11)$$

where X_{CO_2} is equal to the fraction of CO_2 in the bubble. Thus, the mass transfer coefficient reduces with time as a result of the reducing concentration of CO_2 . Changes in bubble diameter, which can be either expansion or shrinking in the case of CO_2/N_2 mixtures, will also cause a change in the mass transfer coefficient.

Hydrate Compensation

The diameter reduction rate for a hydrate coated bubble as reported by Fujioka *et al.* (1994) is used to determine the mass transfer coefficient for pure CO_2 bubbles at depths below 1000 m. Equation 4-5 is used to convert the reduction rate to a molar transfer rate. Solving for change in bubble diameter over time yields:

$$\frac{dD_b}{dt} = 6\alpha_D \frac{\rho_w}{\rho_g} x_{gs} \quad (4-11).$$

Rearranging for α_D yields:

$$\alpha_D = \frac{dD_b}{dt} \frac{\rho_g}{6\rho_w x_{gs}} \quad (4-12).$$

The mass transfer coefficient, α_D , is then converted to a molar transfer rate. As a result

$$\alpha_D = \frac{dD_b}{dt} \frac{\rho_g}{6} \frac{1000}{44.01} \pi D_b^2 \left[\frac{\text{mol } CO_2}{s} \right] \quad (4-13)$$

Ideally, the model would modify the mass transfer coefficient for all CO₂/N₂ mixtures within their hydrate formation region. However, there is insufficient work on the CO₂/N₂ mixtures and hydrate formation to provide the insight necessary to model this behavior.

Chapter 5

5. Simulation Results and Discussion

5.1 CO₂/N₂ Mixed Gas Single Bubble Release

5.1.1 Mass Transfer Coefficient Analysis

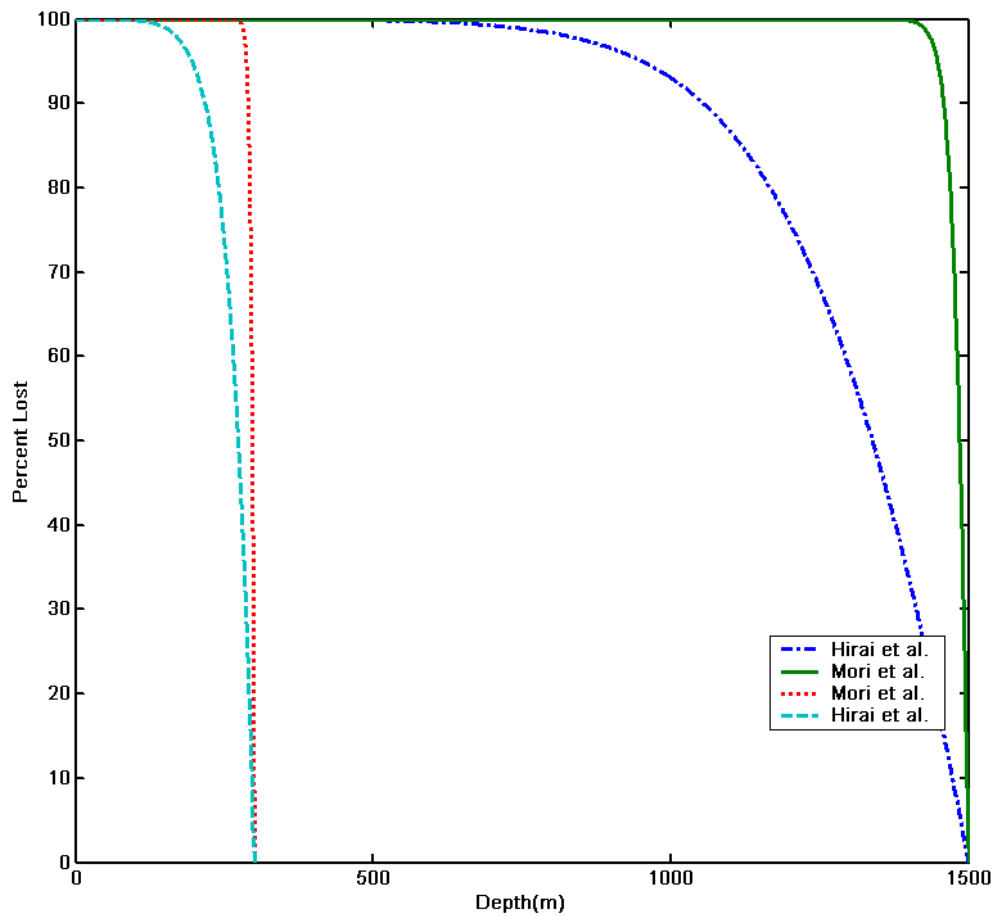


Figure 5-1: Effect of mass transfer coefficient on dissolution of 50% CO₂ bubble

The mass transfer rate as computed following the method suggested by Mori *et al.* predicts much faster dissolution rates than the rate reported by Hirai *et al.* Figure 5-1

shows the difference between predictions for a 50% CO₂, 1 cm bubble at two different injection depths. Figure 5-2 shows the difference between predictions for two different initial concentrations of a 1 cm bubble injected at 500 m. It is seen in the figures that the difference increases with increases in injection depth but remains relatively constant for changes in initial bubble concentrations.

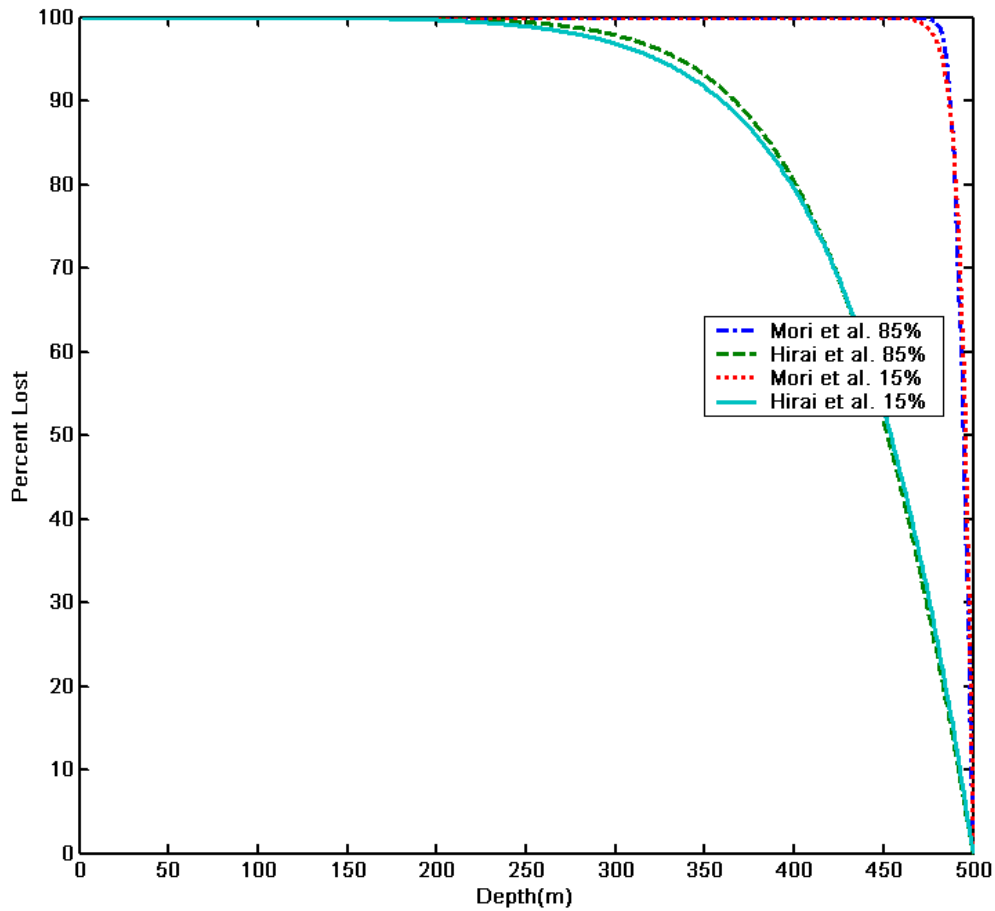


Figure 5-2: Effect of mass transfer coefficient on dissolution of bubbles injected at 500 m

Most experimental work more closely supports the dissolution rates observed by Hirai *et al.* (1996) than those predicted by Mori *et al.* (1998). This discrepancy between the empirical correlations and actual bubble behavior suggest a need for further investigation

into the dissolution rates of bubbles and development of new correlations. Such large differences in predictions and behavior would be costly in applications of a sequestration process. The remaining calculations in this section are performed using the mass transfer rate suggested by Hirai *et al.* (1996). This provides a safe estimate of injection depths required and economic costs based on a slow dissolution rate.

5.1.2 Simulator Output

The mixed gas bubble simulator provides a detailed graphical representation of the behavior of an injected mixed gas bubble. Figures 5-7-5-18 are the simulator output for bubble concentrations of 15%, 50%, 85%, and 100% CO₂ solutions injected at depths of 300, 500, 1000, and 1500m. All bubbles are 1 cm bubbles unless otherwise specified.

The depth versus time plot allows an analysis of the bubble rise velocity by presenting bubble depth over time. The density plot provides insight into the main driving force behind bubble rise, the bubble and seawater density differential. Additionally, it displays the phase behavior of the mixed gas and allows comparison to the phase behavior of pure CO₂. The remaining plots log the dissolution process, presenting the composition, diameter, and mass change in the bubble with depth. Combined, the plots allow a complete analysis of the performance of both the dissolution and transportation processes for any CO₂/N₂ mixed gas bubble injection.

5.13 Rise Velocity

Figures 5-19 - 5-22 display the travel time of different composition bubbles for four different injection depths. Several important trends are noticeable in the plots. Firstly, at greater injection depths decreases in initial bubble CO₂ content increase the bubble rise velocity. This results from the low density and compressibility of N₂. As depth increases the density of CO₂ approaches that of the seawater. The density of N₂ increases with depth, as well, but it does so at a much slower pace. The density plots of the simulator outputs show the growth in the density differential between the mixed gas bubbles and a pure CO₂ bubble with increasing depth. Greater quantities of N₂ result in a greater

density differential between the bubble and the seawater with the effect more predominate at greater depths.

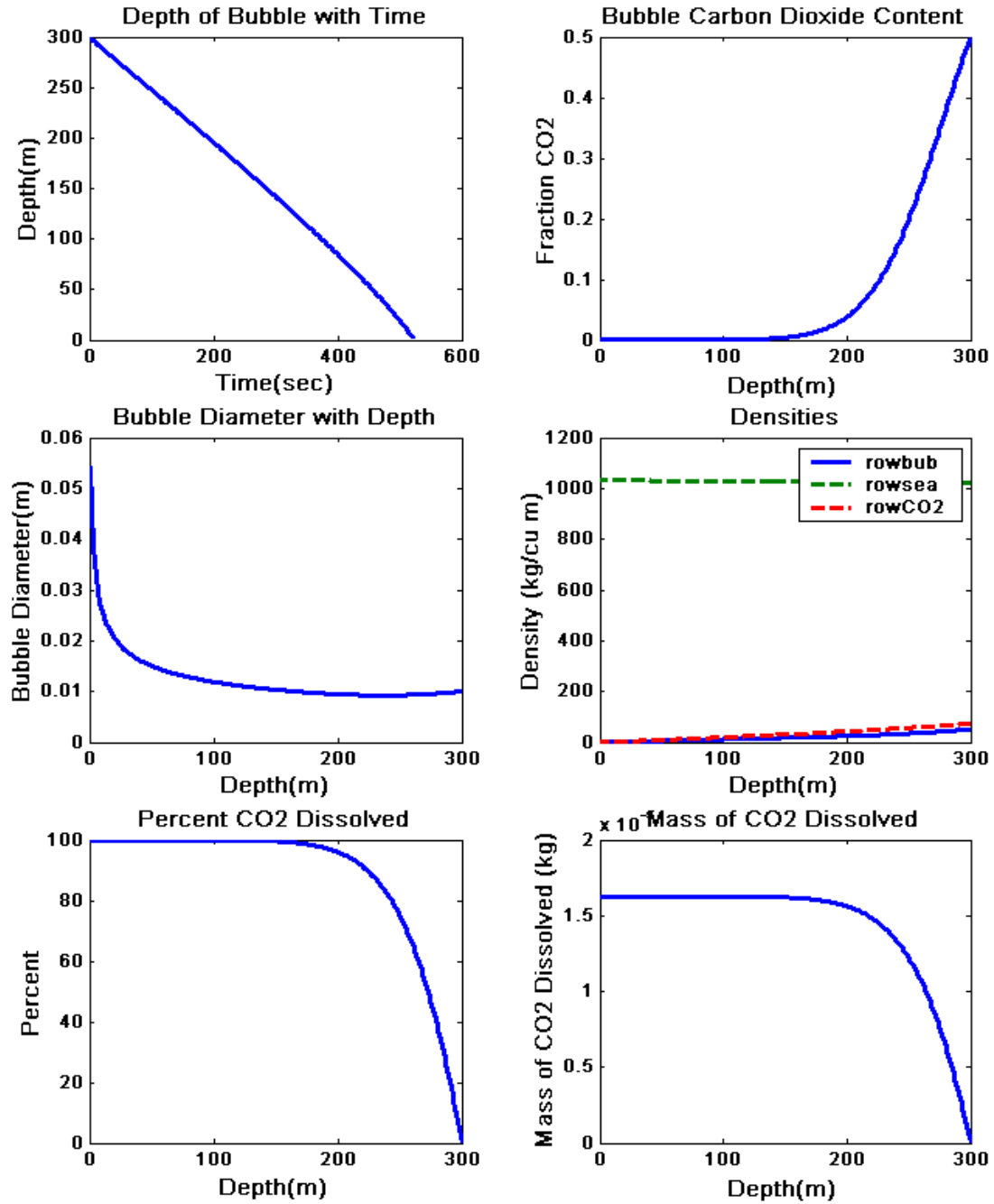


Figure 5-3: Simulator output; 1 cm bubble of 15% CO₂ injected at 300 m

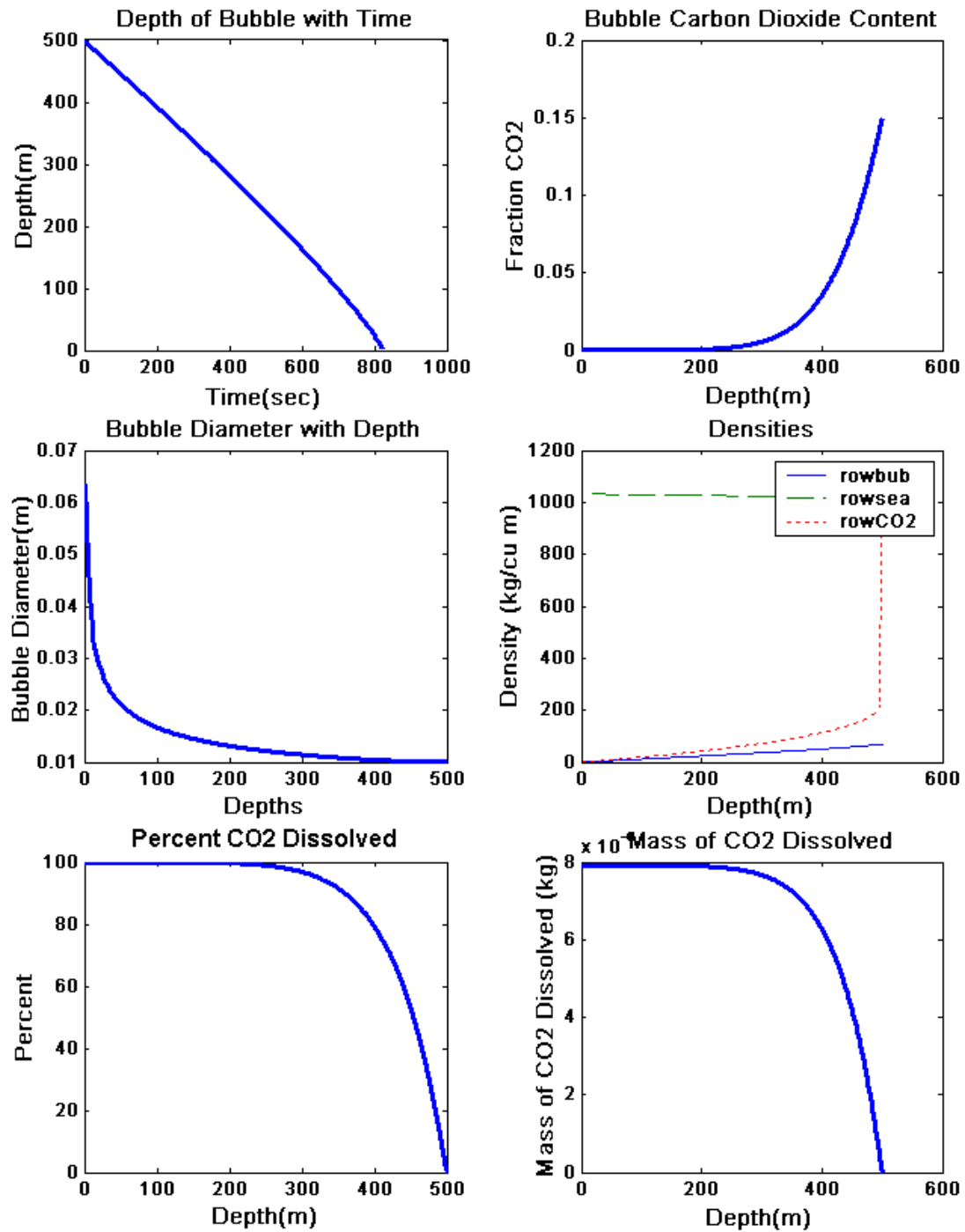


Figure 5-4: Simulator output; 1 cm bubble of 15% CO₂ injected at 500 m

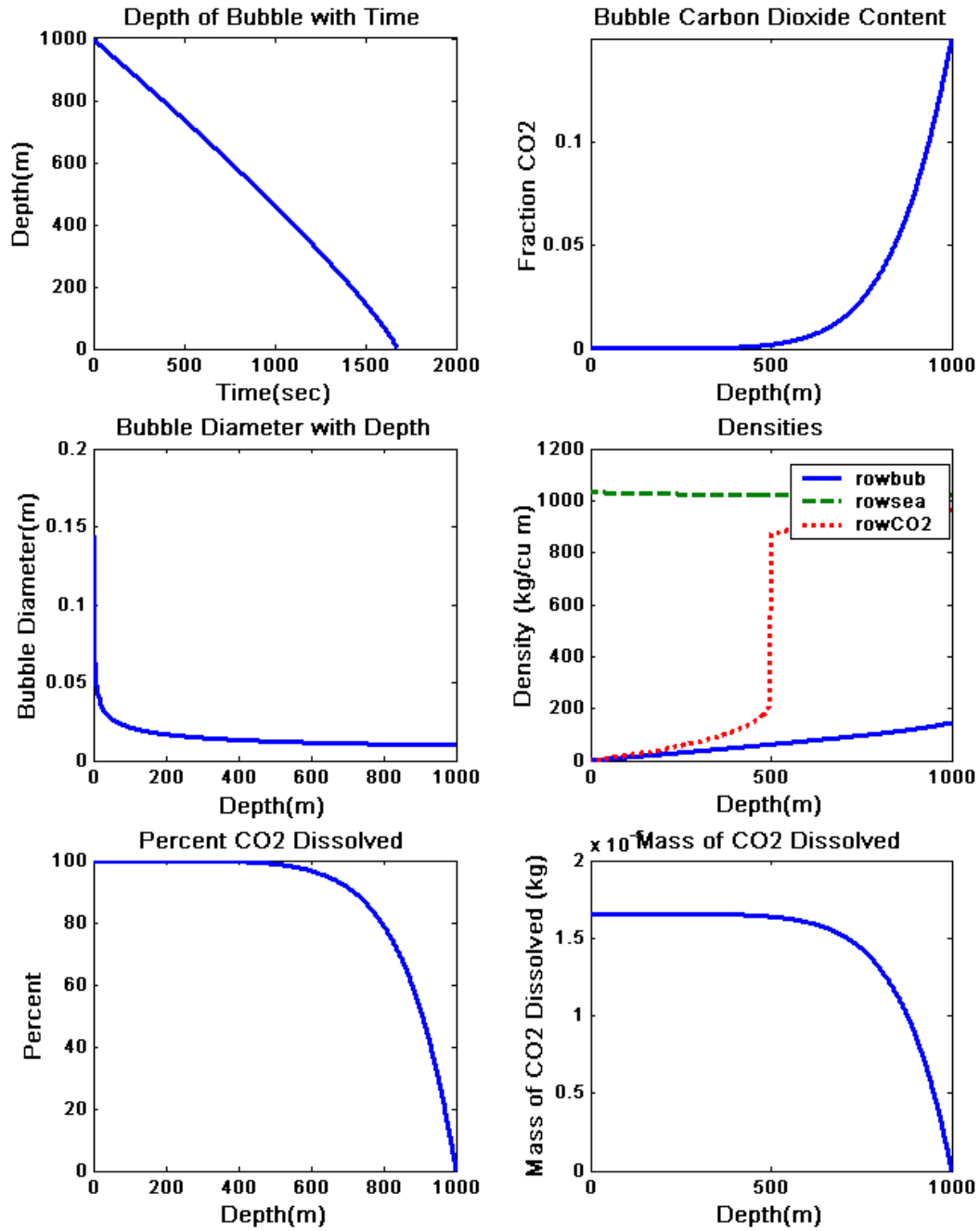


Figure 5-5: Simulator output, 1 cm bubble of 15% CO₂ injected at 1000 m

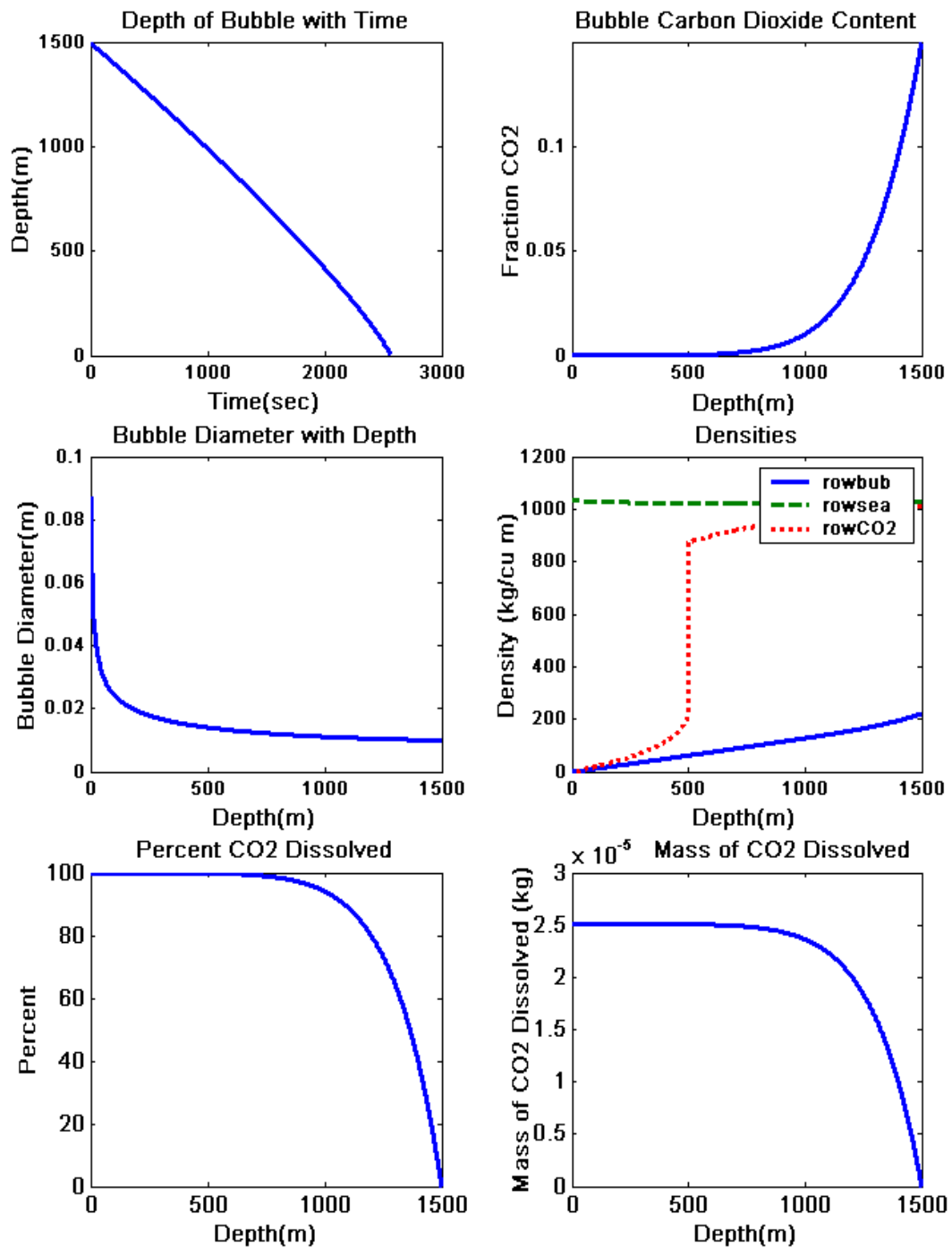


Figure 5-6: Simulator output, 1 cm bubble of 15% CO₂ injected at 1500 m

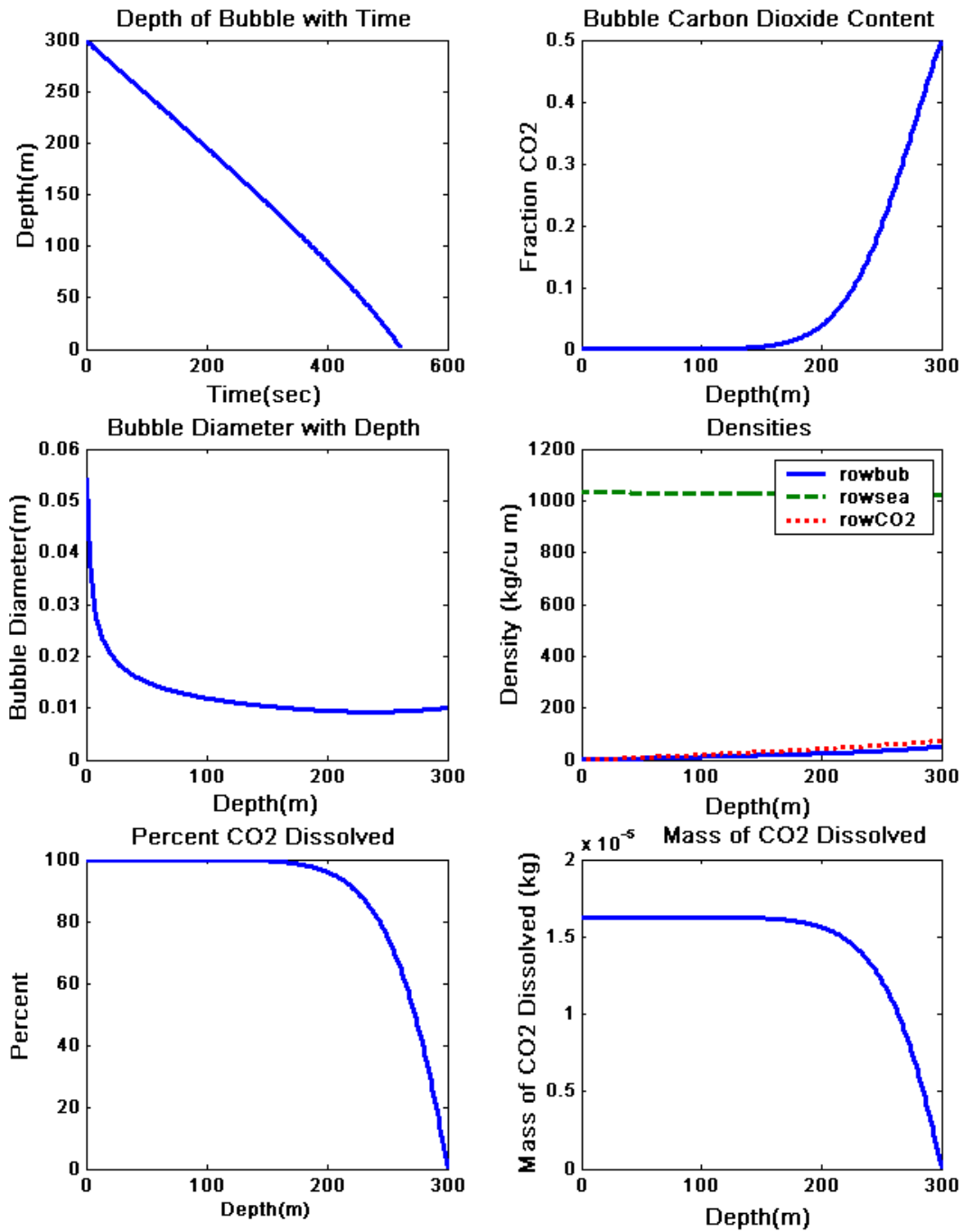


Figure 5-7: Simulator output, 1 cm bubble of 50% CO₂ injected at 300 m

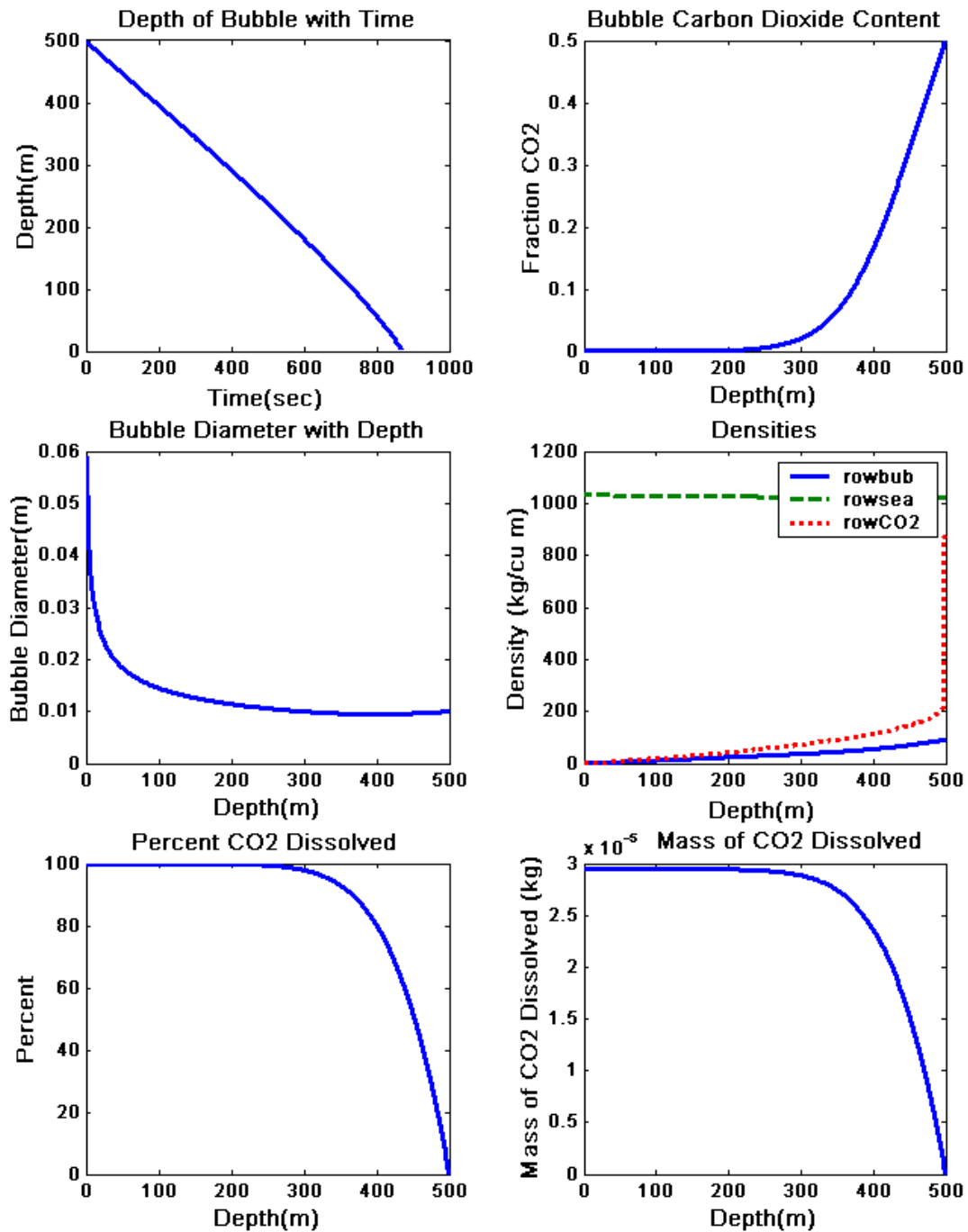


Figure 5-8: Simulator output, 1 cm bubble of 50% CO₂ injected at 500 m

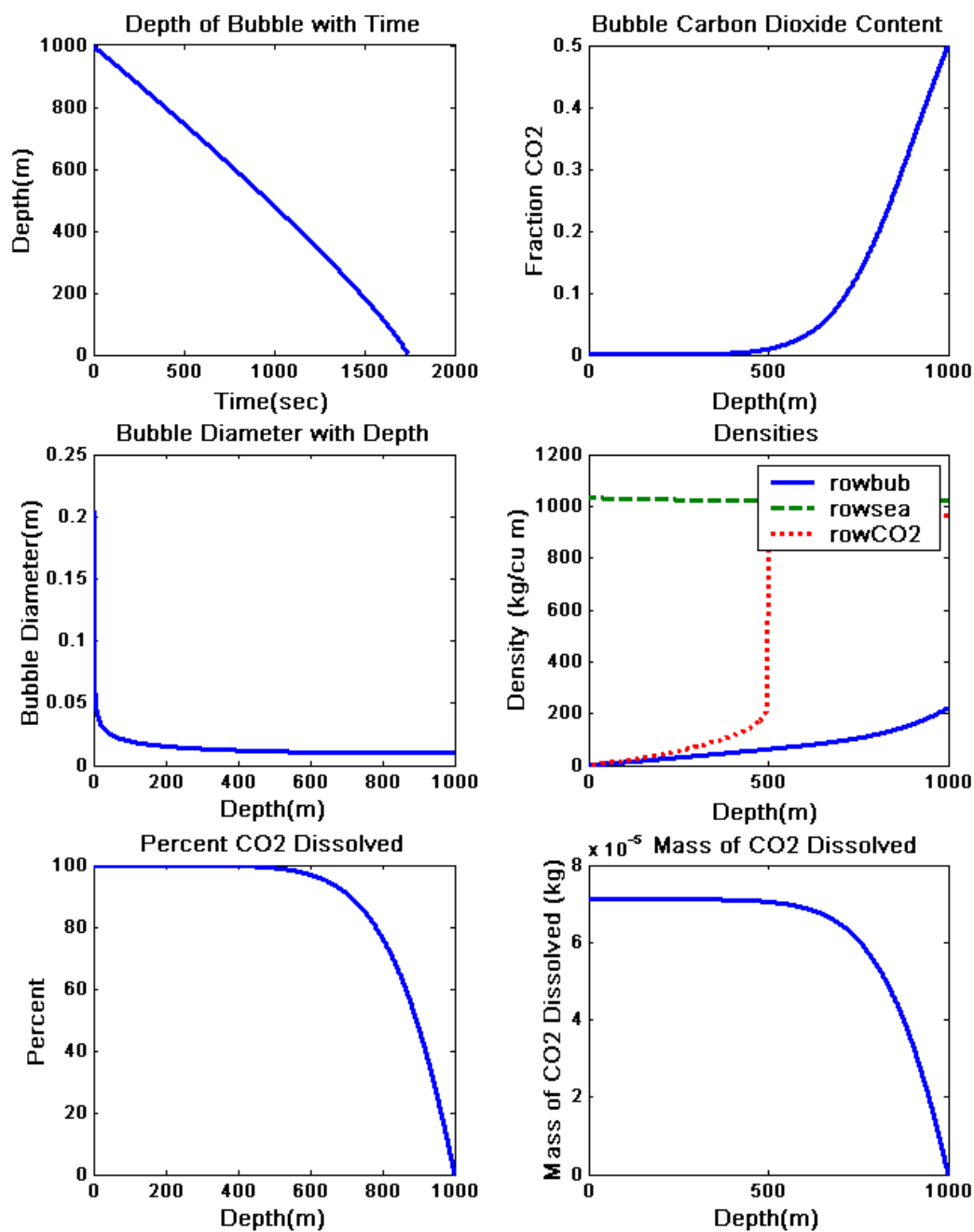


Figure 5-9: Simulator output, 1 cm bubble of 50% CO₂ injected at 1000 m

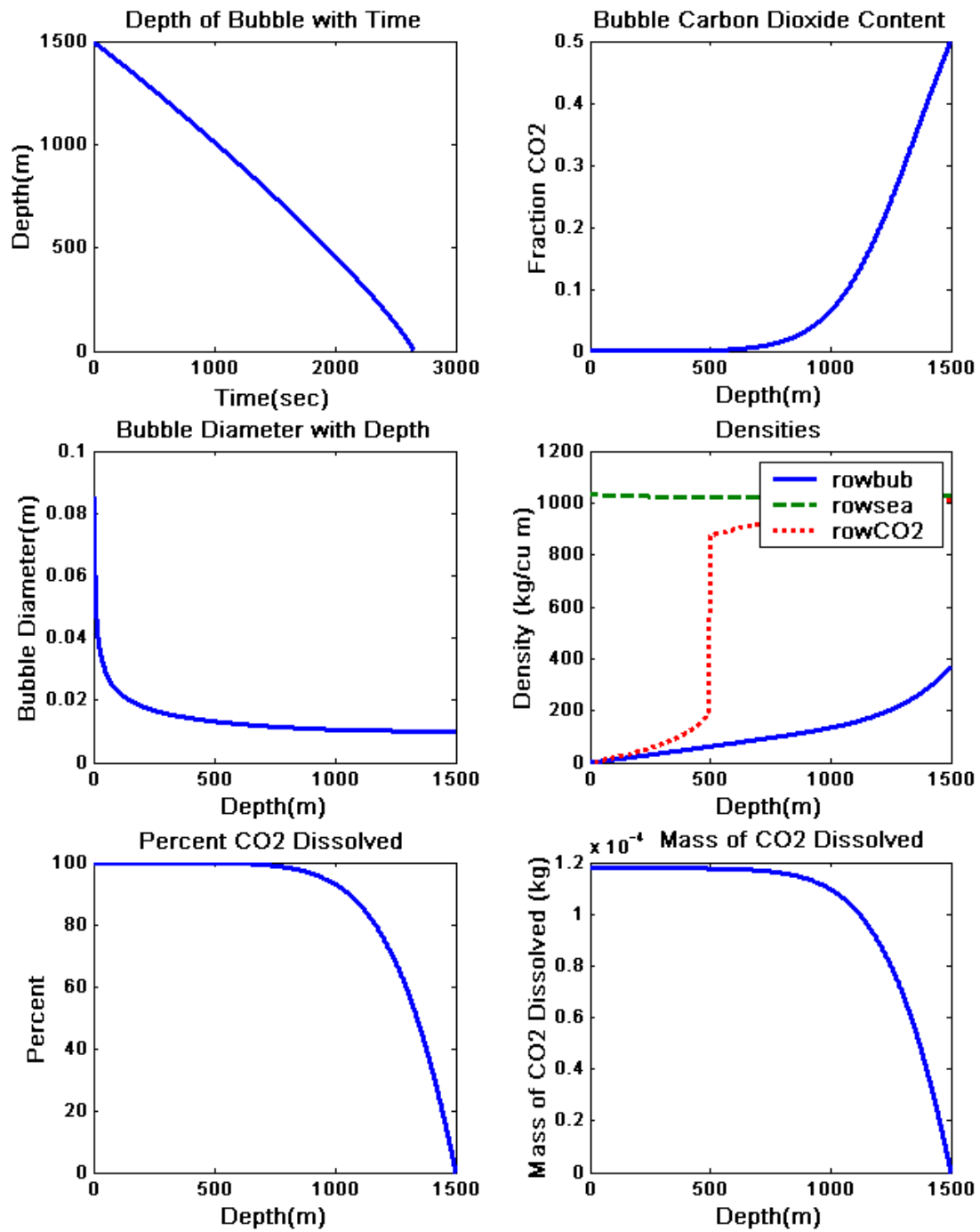


Figure 5-10: Simulator output, 1 cm bubble of 50% CO₂ injected at 1500 m

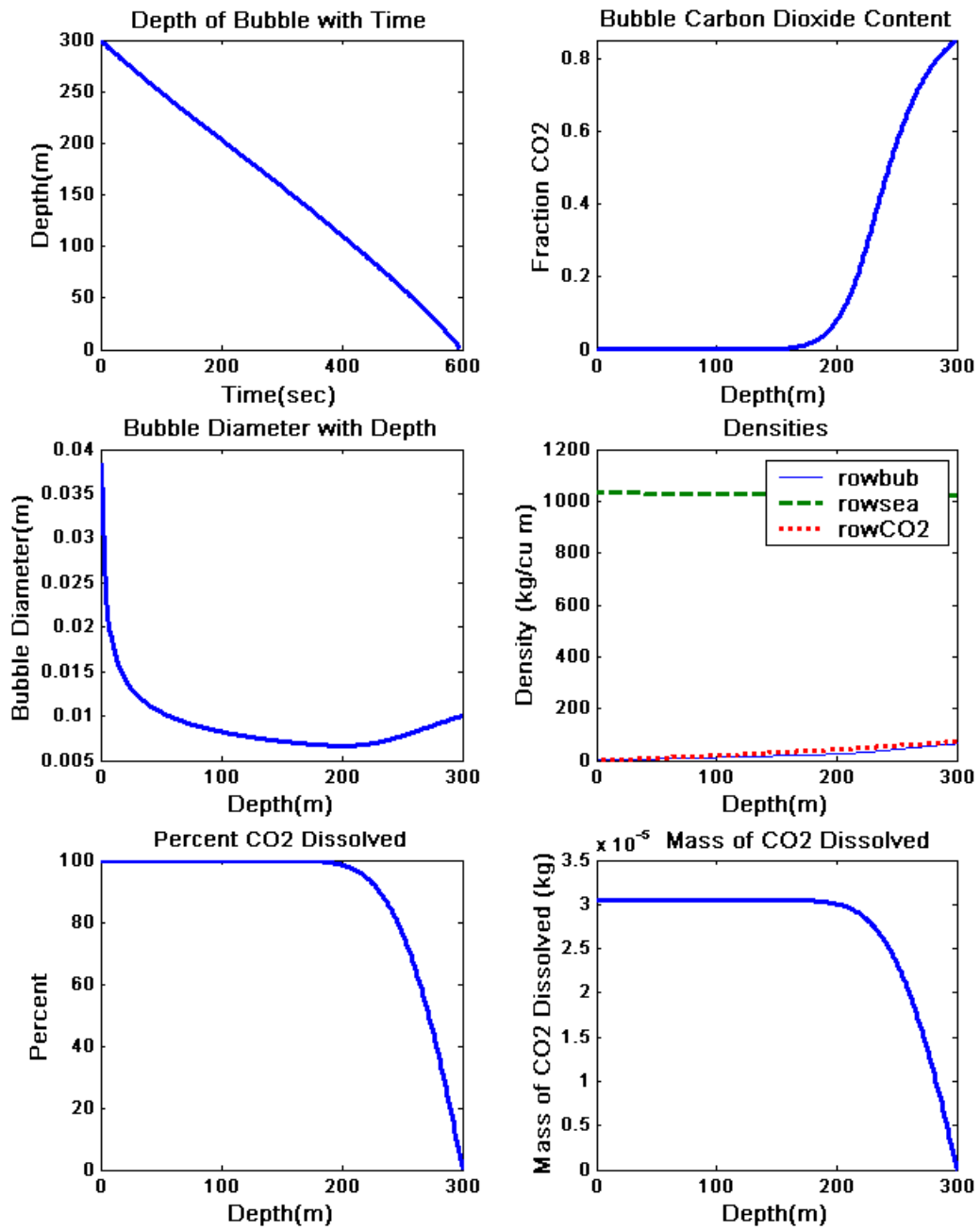


Figure 5-11: Simulator output, 1 cm bubble of 85% CO₂ injected at 300 m

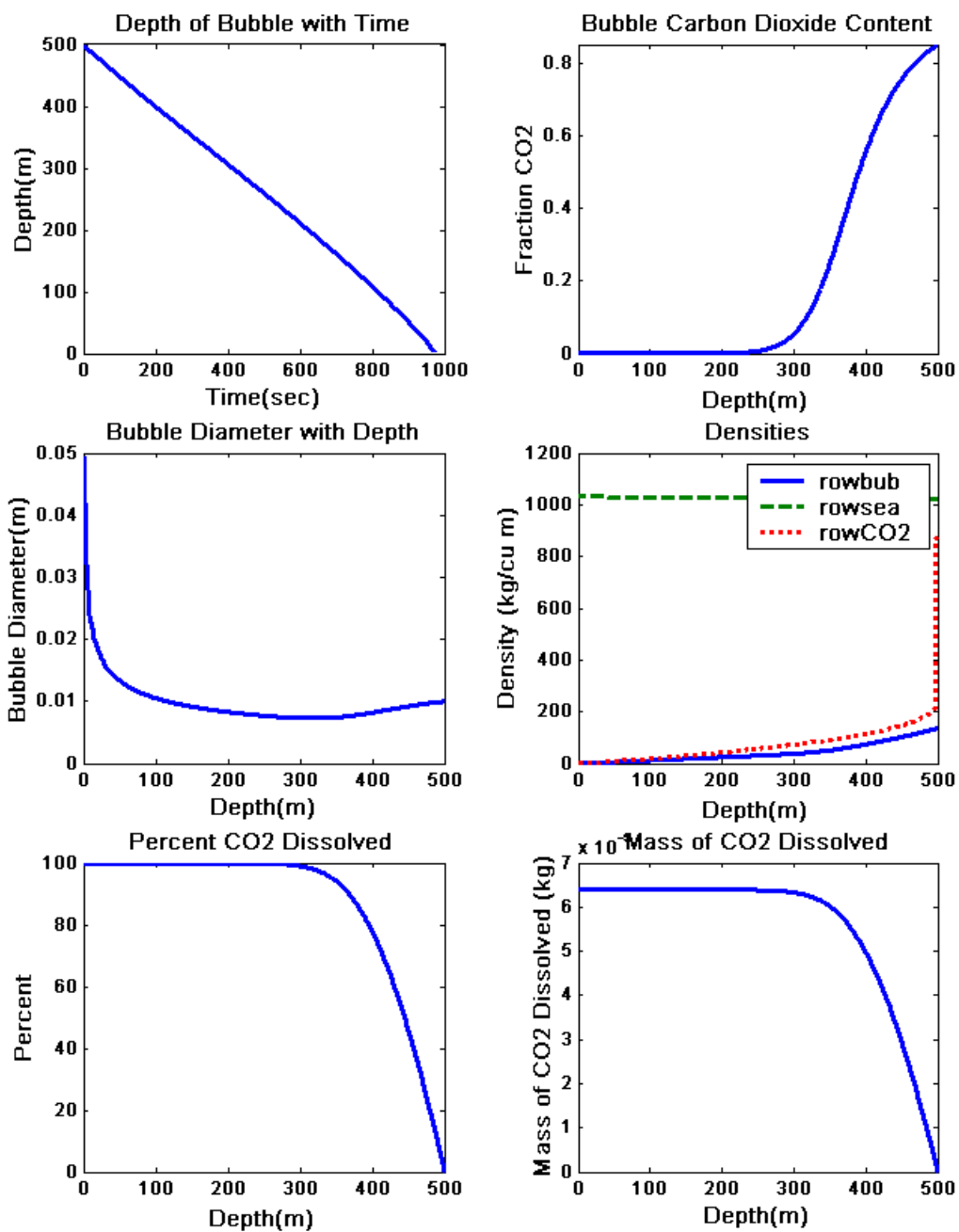


Figure 5-12: Simulator output, 1 cm bubble of 85% CO₂ injected at 500 m

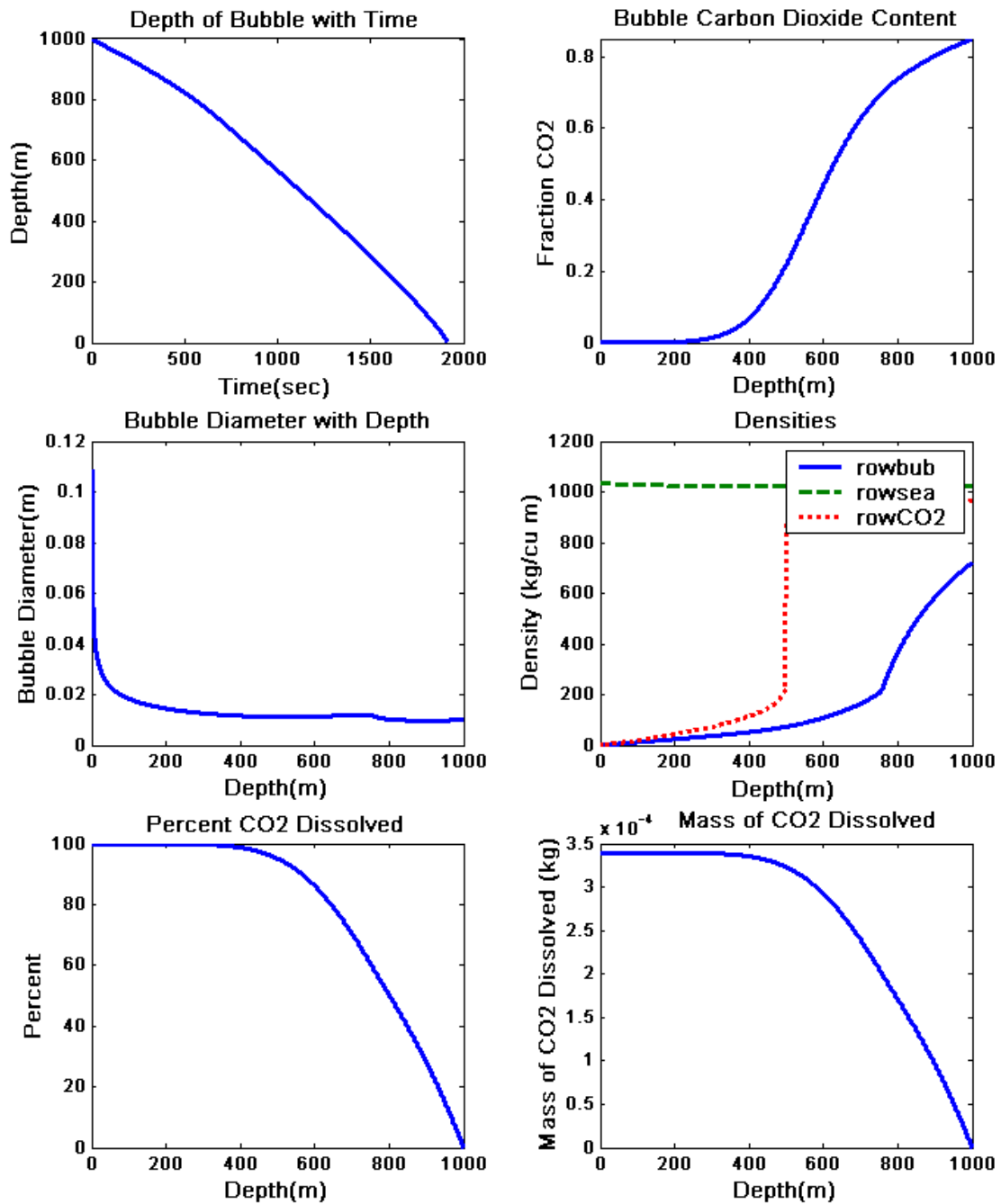


Figure 5-13: Simulator output, 1 cm bubble of 85% CO₂ injected at 1000 m

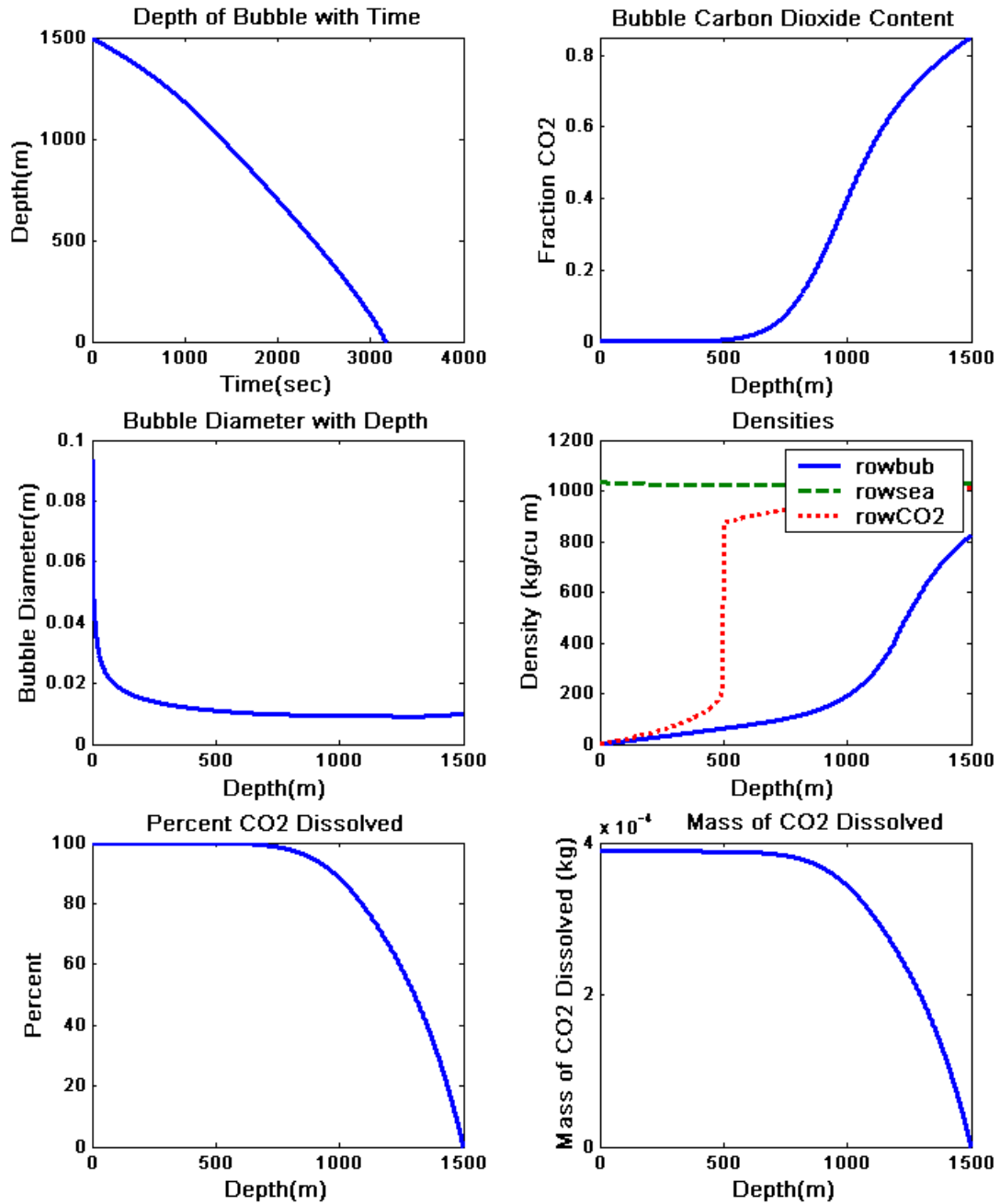


Figure 5-14: Simulator output, 1 cm bubble of 85% CO₂ injected at 1500 m

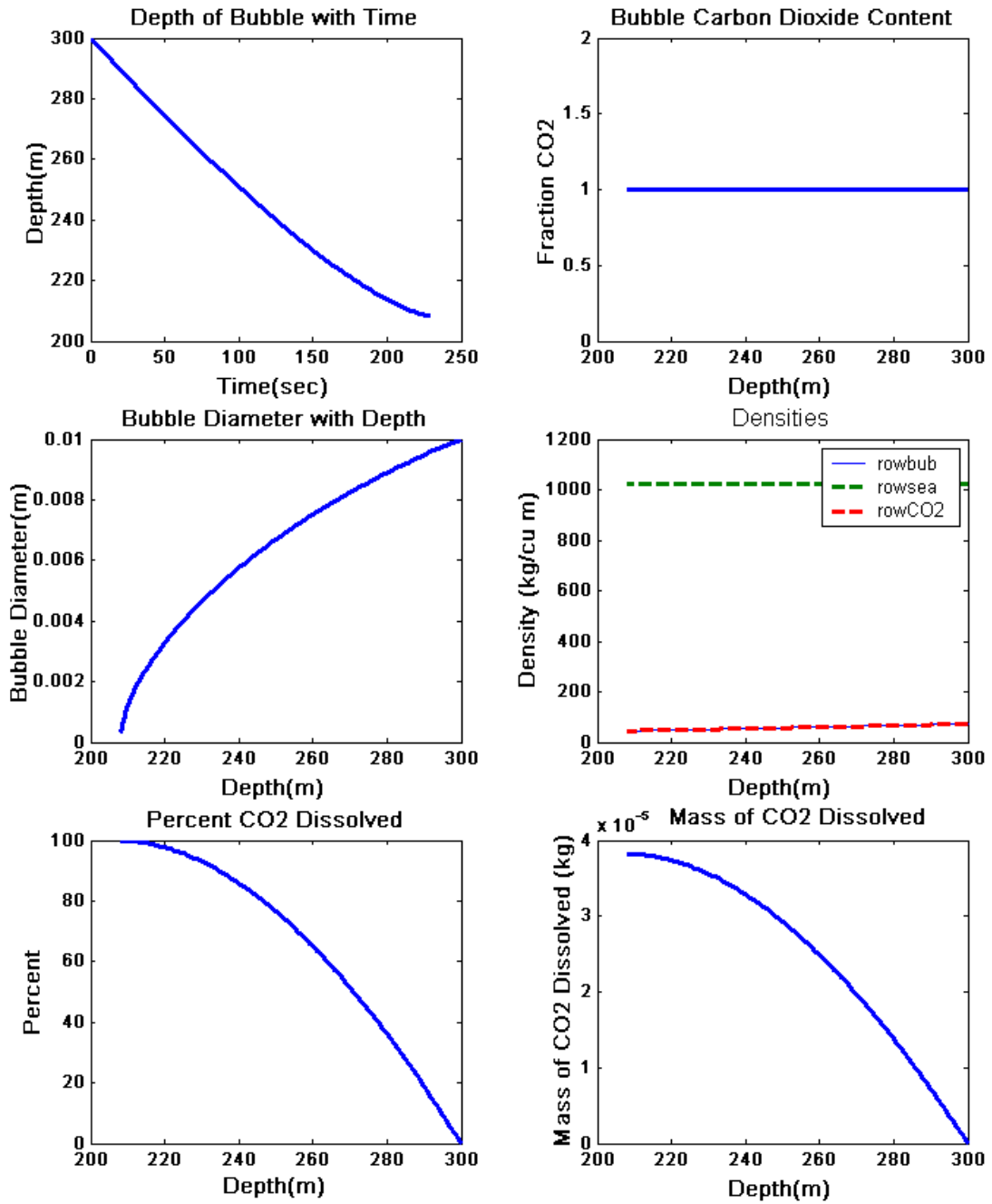


Figure 5-15: Simulator output, 1 cm bubble of 100% CO₂ injected at 300 m

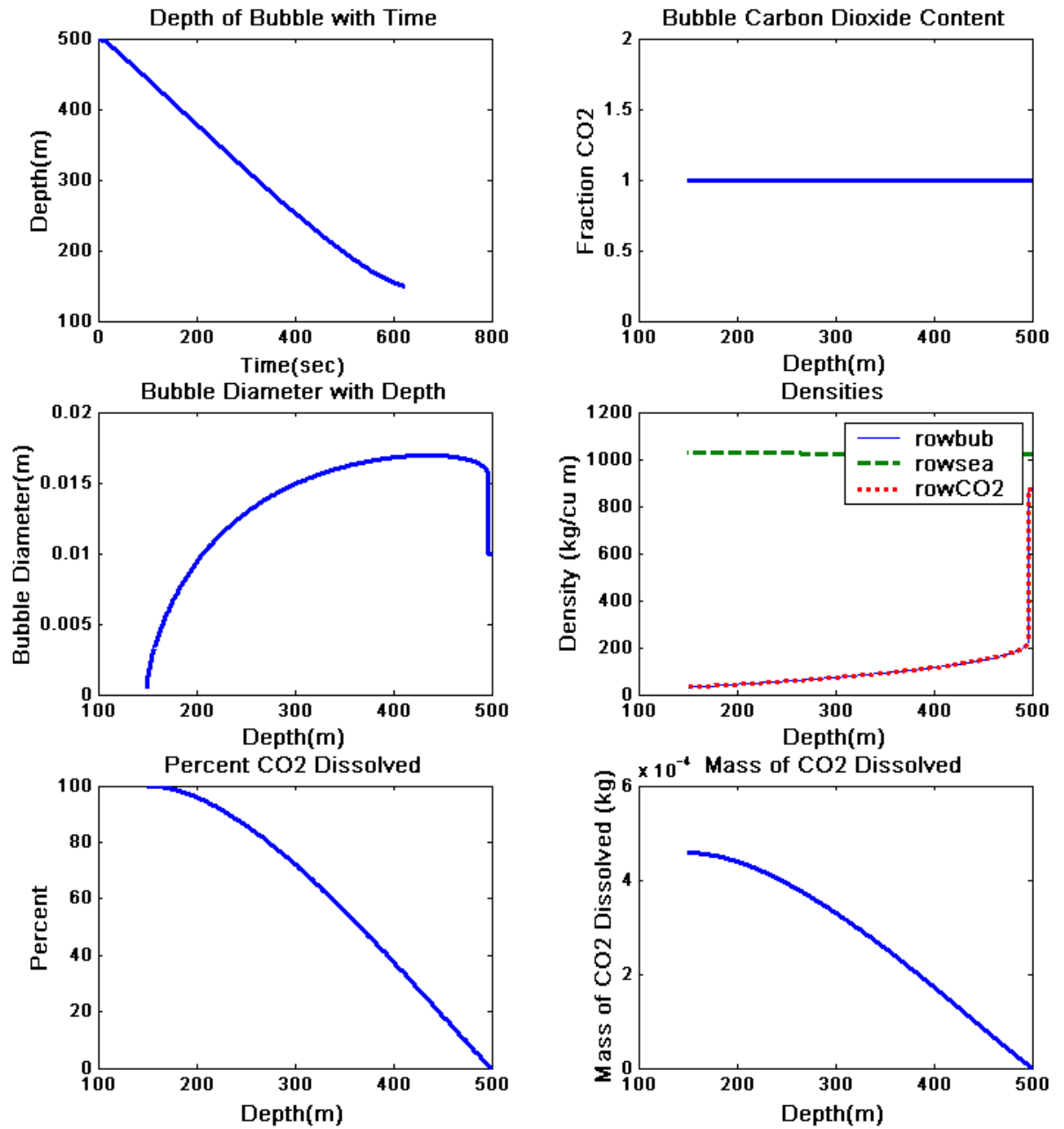


Figure 5-16: Simulator output, 1 cm bubble of 100% CO₂ injected at 500 m

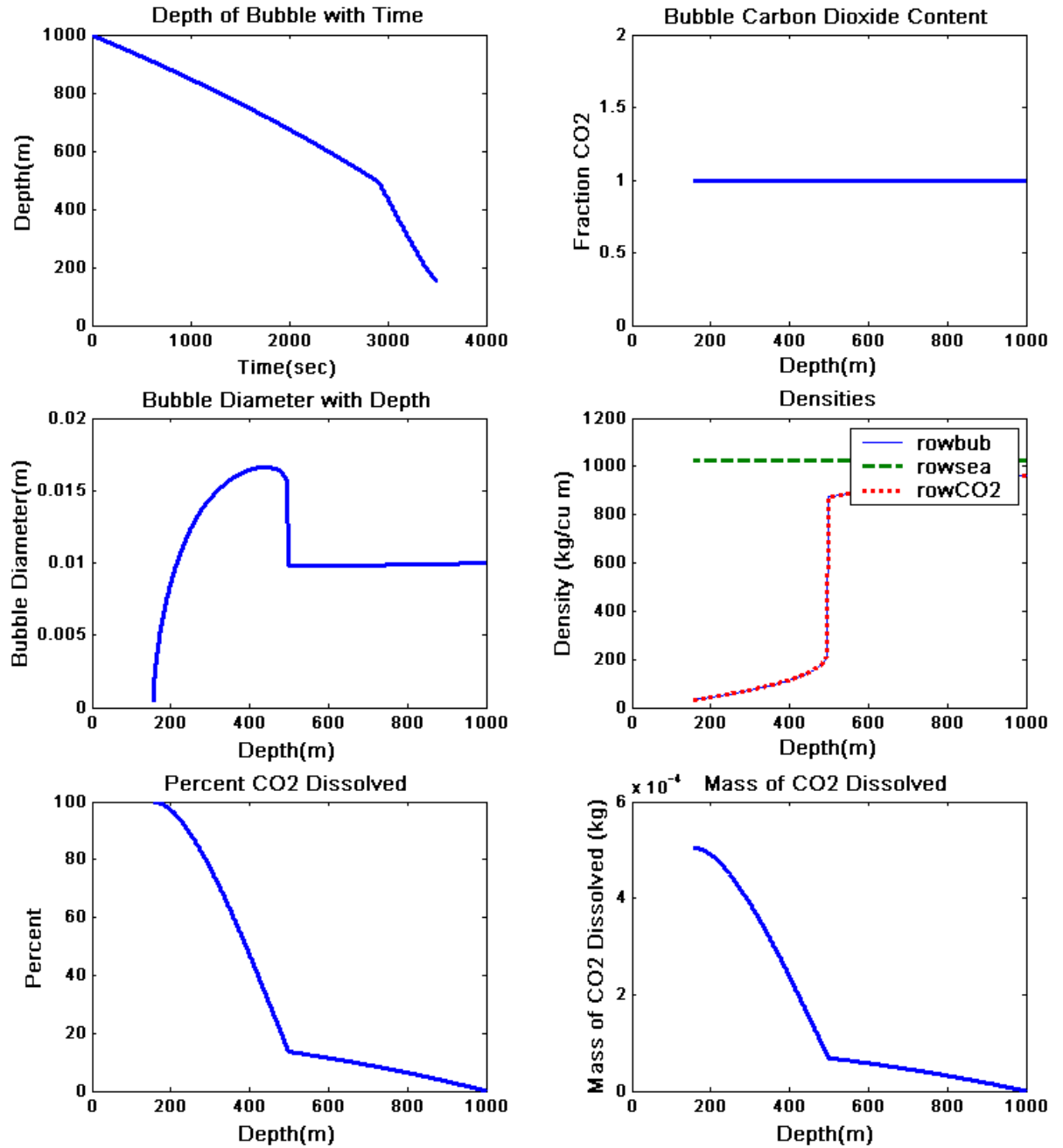


Figure 5-17: Simulator output, 1 cm bubble of 100% CO₂ injected at 1000 m

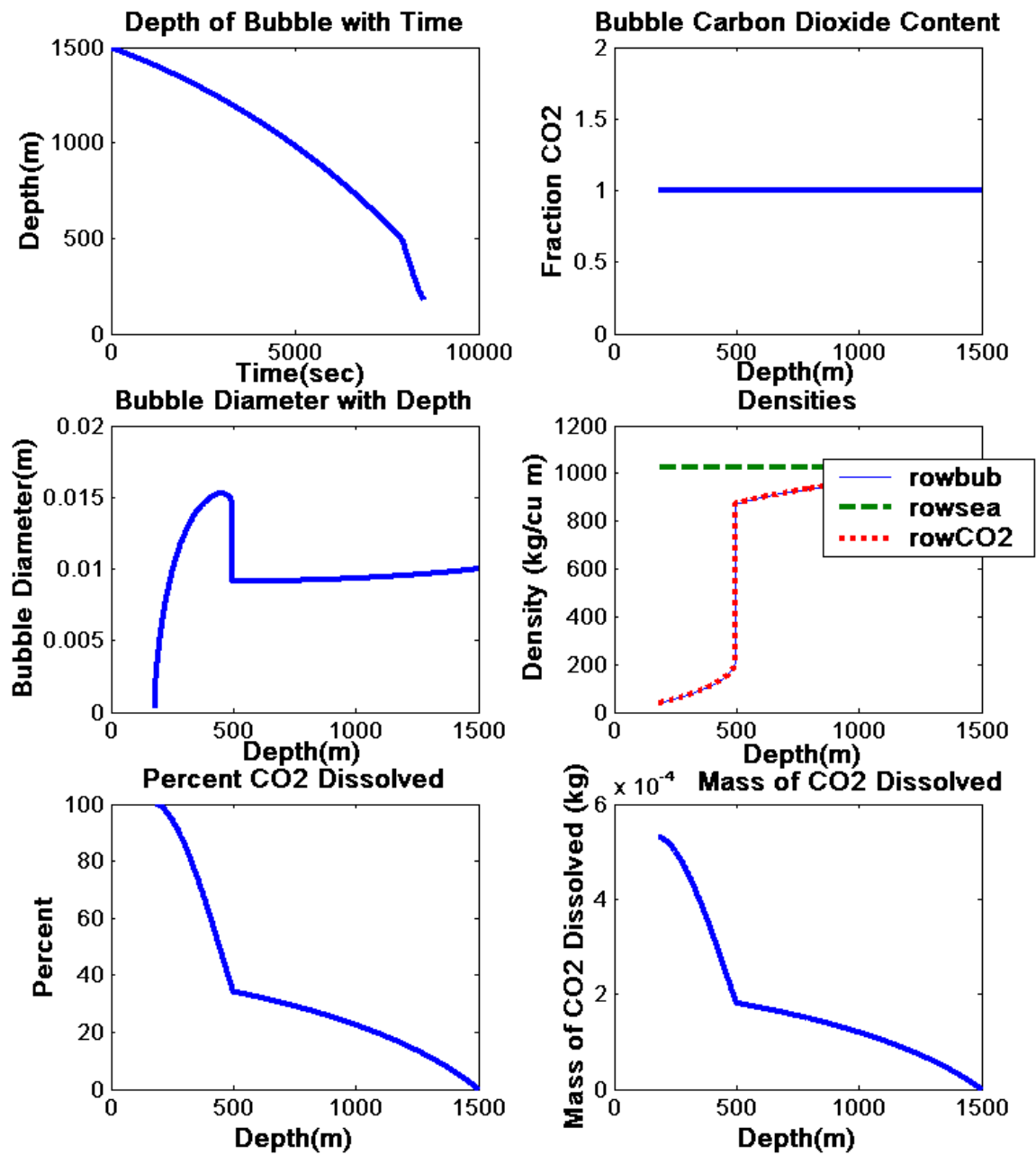


Figure 5-18: Simulator output, 1 cm bubble of 100% CO₂ injected at 1500 m

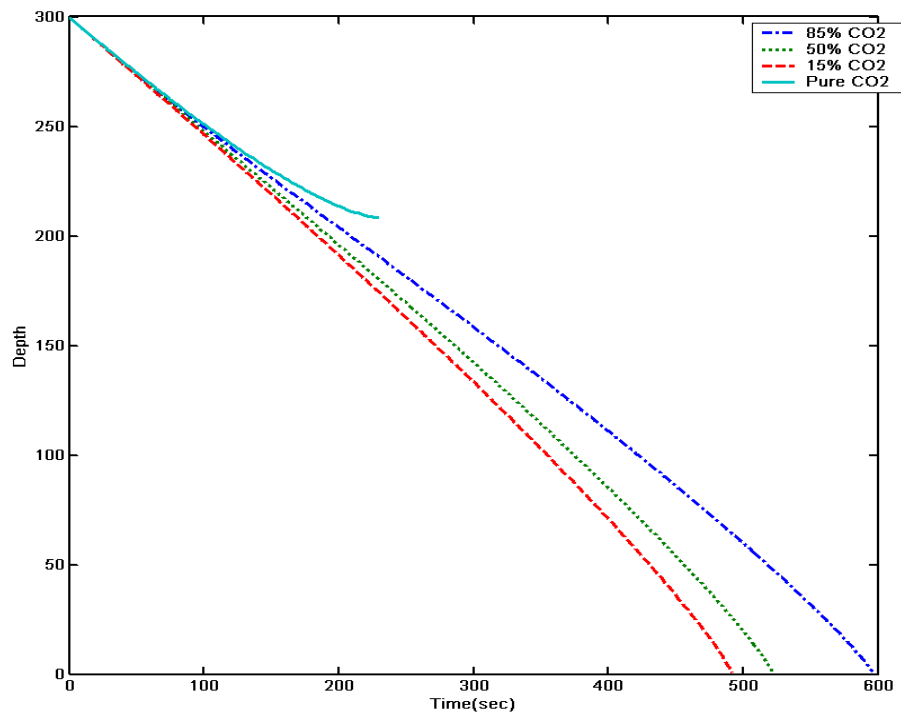


Figure 5-19: Travel time of bubble injected at 300m

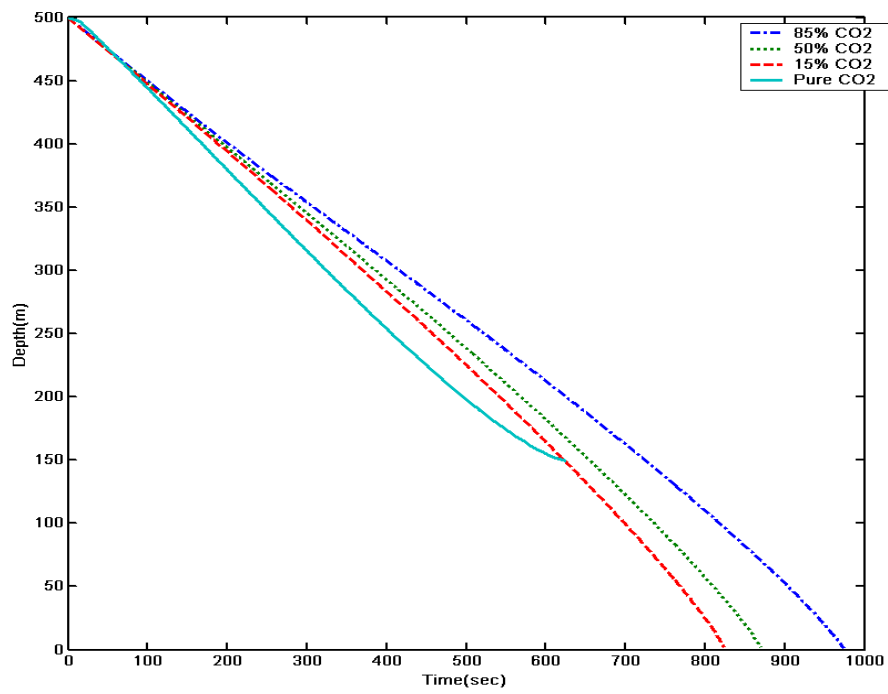


Figure 5-20: Travel time of bubble injected 500m

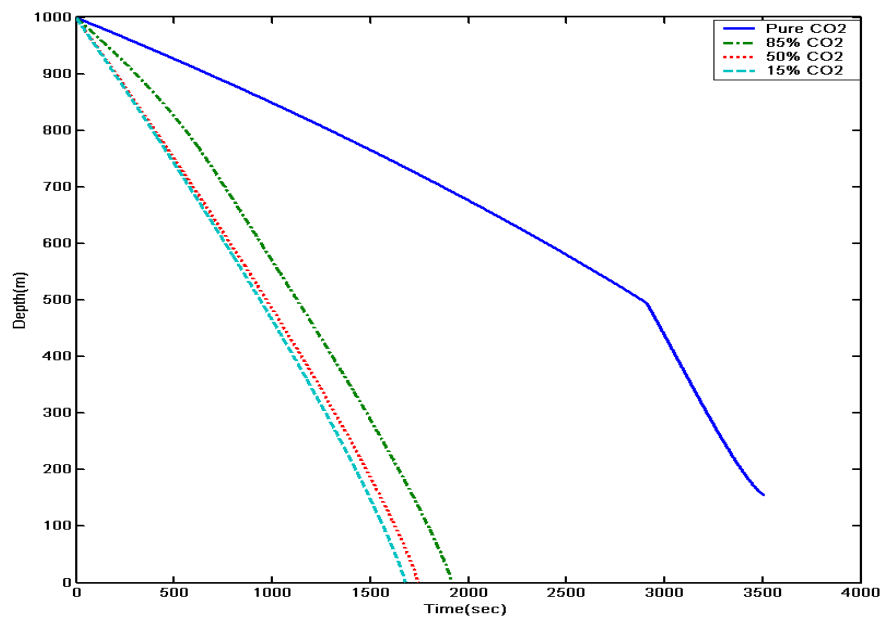


Figure 5-21: Travel time of bubble injected at 1000m

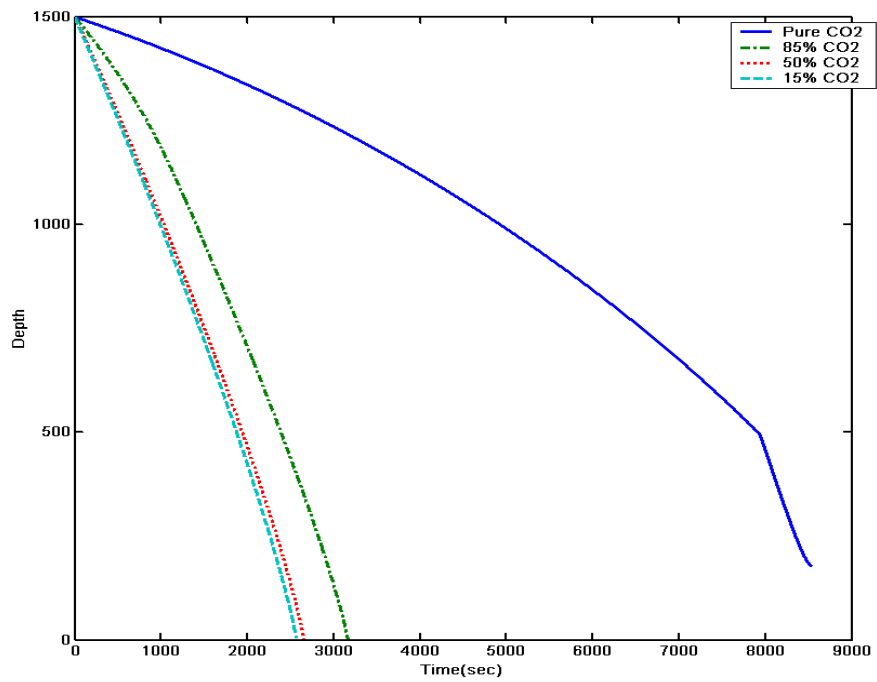


Figure 5-22: Travel time of bubble injected at 1500m

Another important note is the drastic change in bubble rise velocity of the pure CO₂ bubble at 500 m. This depth corresponds to the depth at which the pure CO₂ bubble undergoes a phase change and rapid expansion. The bubble diameter is a parameter in the velocity, both in the calculation of the Reynolds number and the overall drag force. As diameter increases, such as occurs when the pure CO₂ bubble changes from gas to liquid, the rise velocity increases. This is supported by the behavior of the mixed gas bubbles. The N₂ content in the bubbles stays constant. Therefore as the CO₂ content decreases and the bubbles rise, their volume increases both due to increase N₂ percentage and decreased pressure. Consequently, the mixed gas bubbles accelerate as they rise. The pure CO₂ bubble, on the other hand, show a rapid deceleration when their bubble diameter goes to zero.

5.1.4 CO₂ Dissolution

Figures 5-23 – 5-26 graph the percent of CO₂ dissolved with depth for various injection compositions and depths. When injected at 300 m, initial bubble composition exhibits little impact on the dissolution of CO₂ as measured by percentage of initial composition. At least 95% percent of the initial CO₂ of all bubbles is dissolved by the time the bubbles reach the upper mixed layer. As injection depth increases to 500 m, as depicted in Figure 5-24, the dissolution rate of the pure CO₂ falls below that of the mixtures. By 325 m 95% of the CO₂ in all the mixtures has dissolved but only 65% of the pure CO₂ bubble has been dissolved. Figures 5-8 and 5-9 suggest that at lower injection depths, a lower initial CO₂ concentration results to faster dissolution relative to initial CO₂ concentration. Despite the model's failure to account for the possible formation of hydrates by the 85% and 50% CO₂ bubble when injected at 1500 m, the 15% CO₂ bubble still dissolves 90% of initial CO₂ faster than all other bubbles.

The behavior of the pure CO₂ bubble in Figures 5-25 and 5-26 also demonstrate the affect of hydrate formation on CO₂ dissolution. The bubbles show a clear decrease in dissolution rate at depth below 500 meters, where hydrate forms.

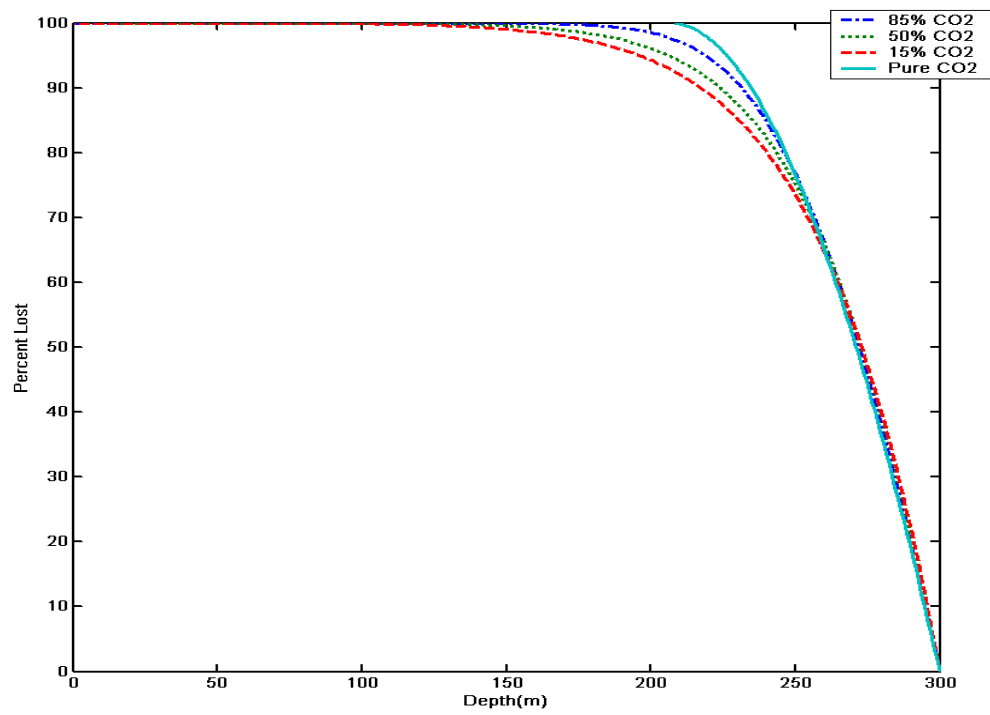


Figure 5-23: Percent CO₂ lost with depth when injected at 300 m

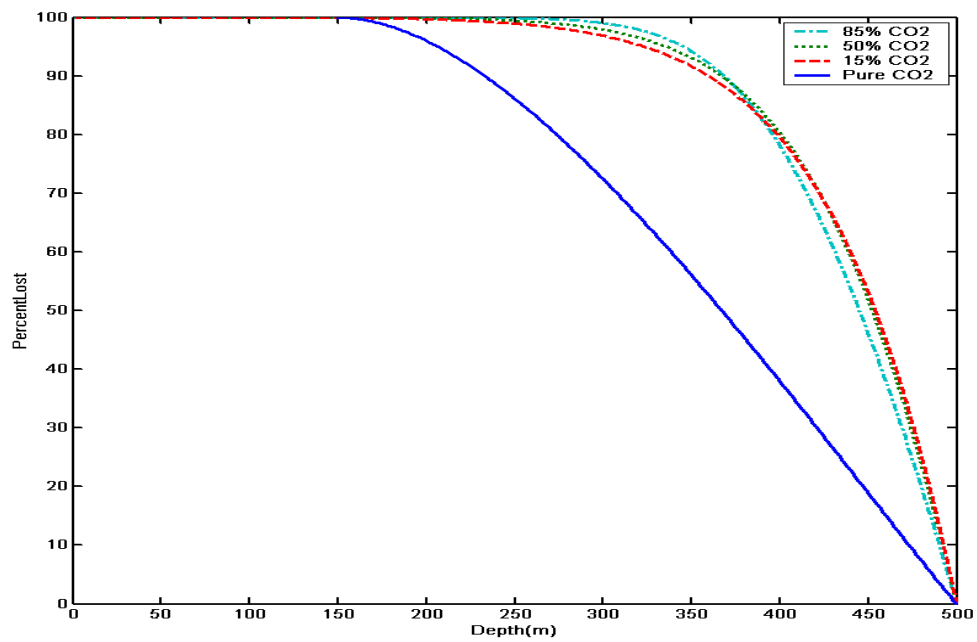


Figure 5-24: Percent CO₂ lost with depth when injected at 500 m

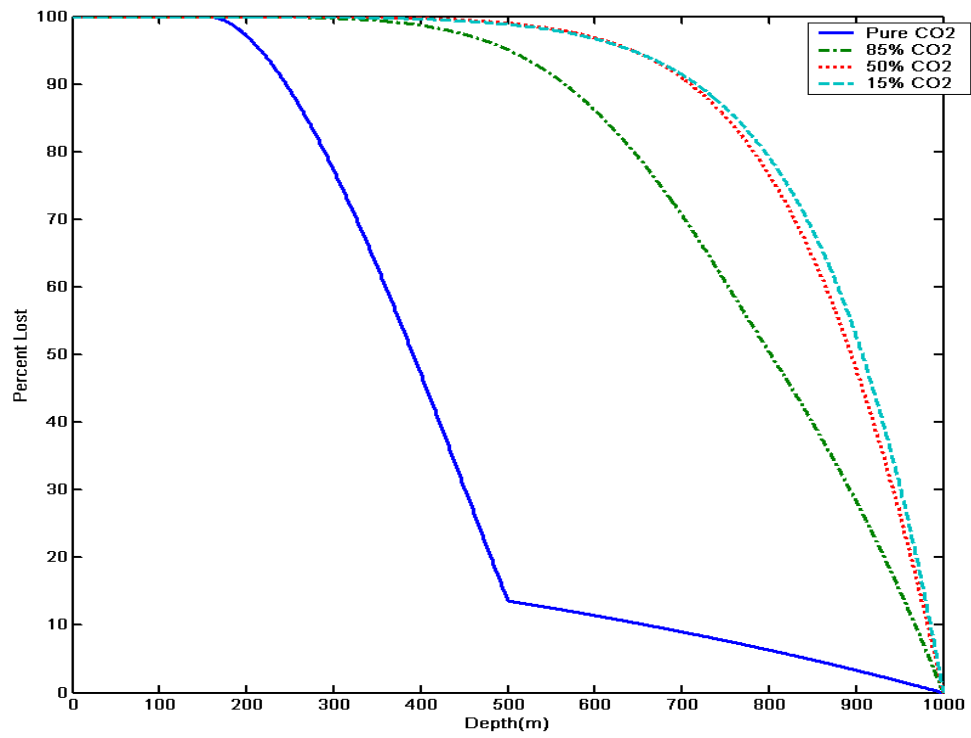


Figure 5-25: Percent CO₂ lost with depth when injected at 1000 m

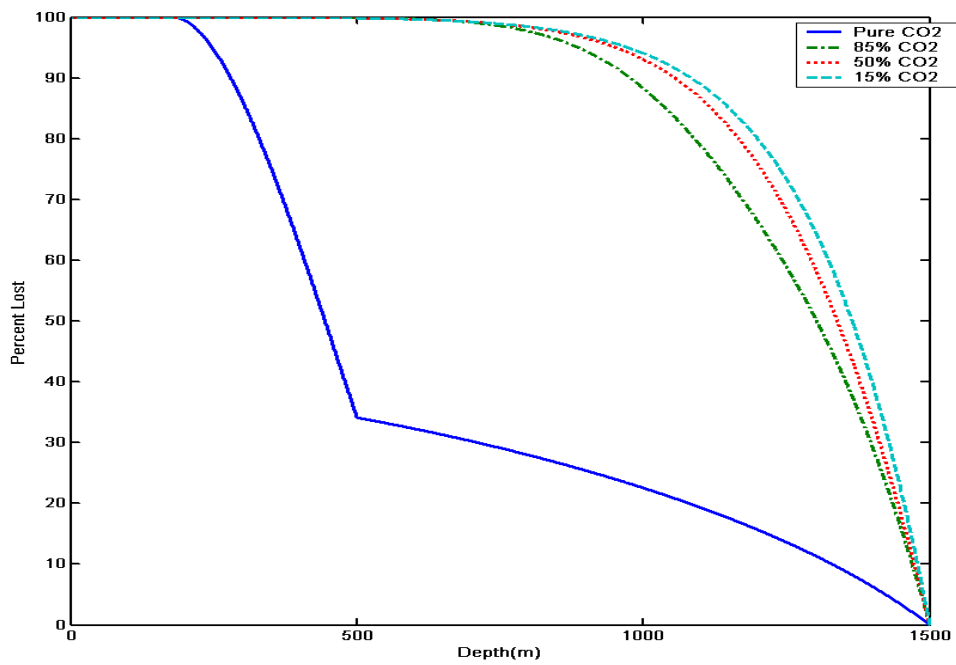


Figure 5-26: Percent CO₂ lost with depth when injected at 1500 m

It is important to consider not only the percentage of CO₂ within a bubble that dissolves, but also the total mass of CO₂ that dissolves. 90% of the CO₂ in a bubble that is only initially 15% CO₂, is a much smaller mass than 90% of a pure CO₂ bubble. Figure 5-27 plots the total mass of CO₂ dissolved for various initial compositions injected at 1500 m. The plot emphasizes the importance of including hydrate behavior in a sequestration model. The 85% CO₂ bubble dissolves a larger mass of CO₂ in depths below approximately 400 m. This implies that an 85% CO₂ mixture will have greater success at long term CO₂ sequestration than pure CO₂ when injected at 1500m. Of course, this depends on whether or not the 85% mixture itself forms hydrate. A second noteworthy feature of Figure 5-10 is the low mass of CO₂ dissolved when the initial injection concentration is 50% or below. The volume of an injected gas necessary to dissolve a given mass of CO₂ increases with an increase in the initial impurity level of that gas.

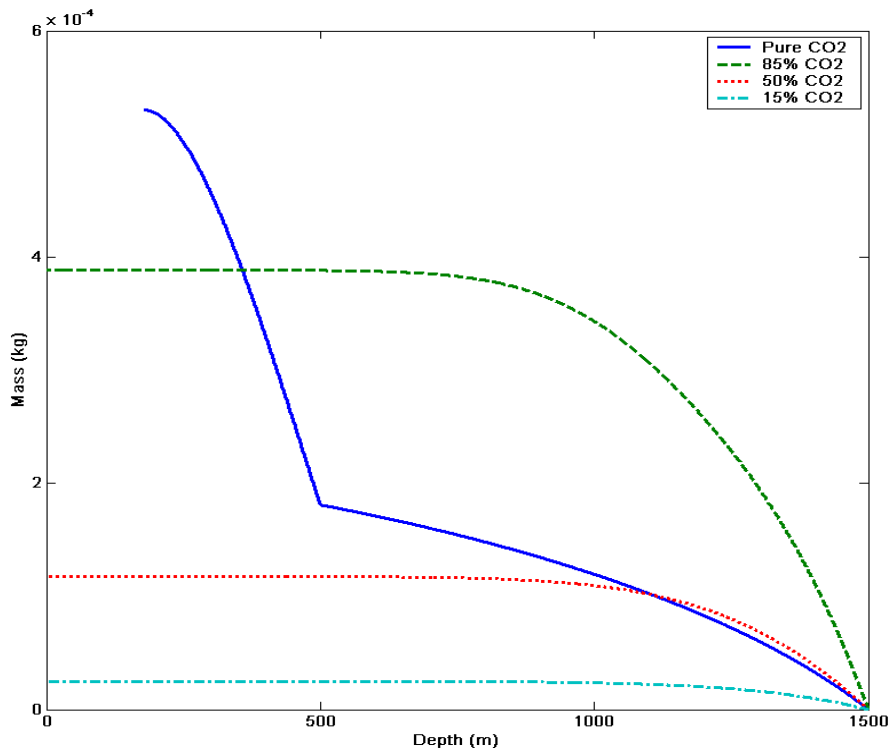


Figure 5-27: Mass of CO₂ dissolved with depth when a 1 cm bubble is injected at 1500m

5.1.5 Variations in Bubble Diameter

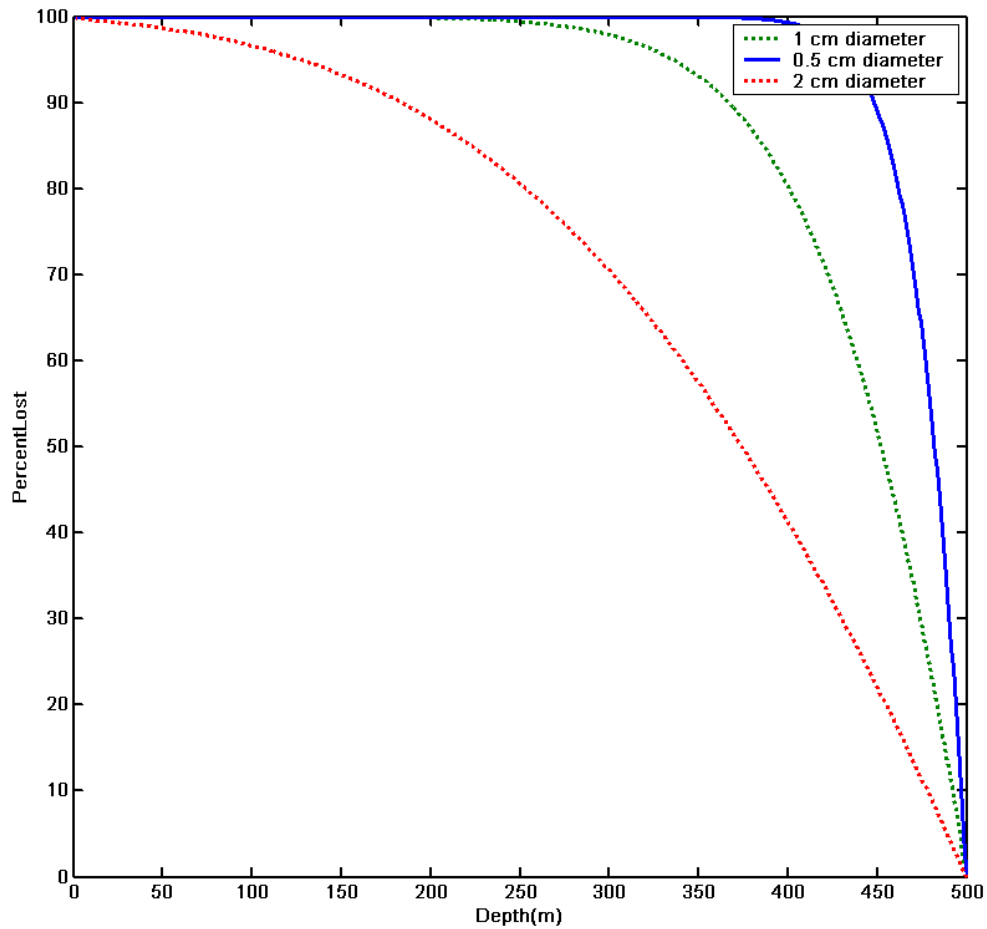


Figure 5-28: Effect of bubble diameter on dissolution rate of 50% CO₂ bubble injected at 500 m

Figure 5-28 plots the percent of CO₂ lost for a 1 cm 50% bubble with depth at various initial bubble diameters. The plot shows a distinct decrease in dissolution rate with an increase in bubble diameter. From Figure 5-29 it can be seen that bubble diameter has the opposite effect on bubble rise velocities. As the bubble diameter increases, the bubble rise velocity increases as well. From these plots it is apparent that bubble diameter is a key parameter in bubble dissolution and velocity.

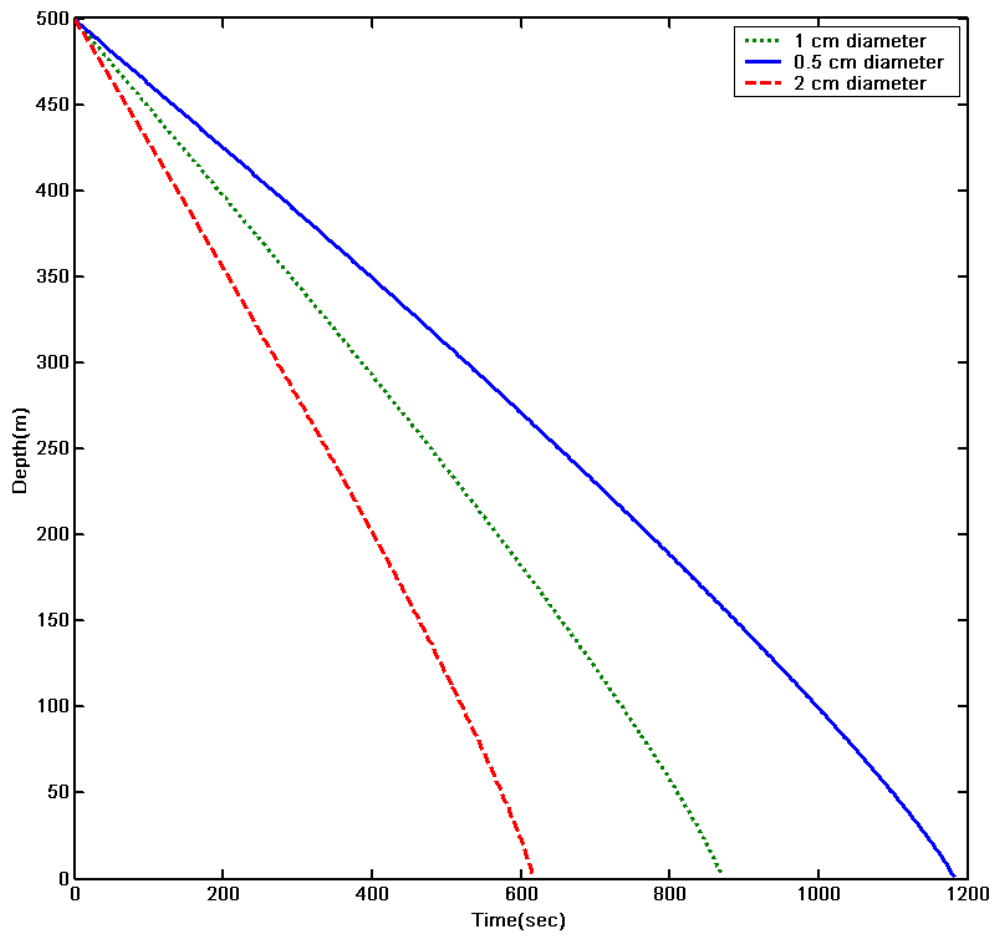


Figure 5-29: Effect of bubble diameter on rise rate; 50% CO₂ bubble injected at 500 m

5.1 P-GLAD Simulation

5.2.1 Water Flow Rates

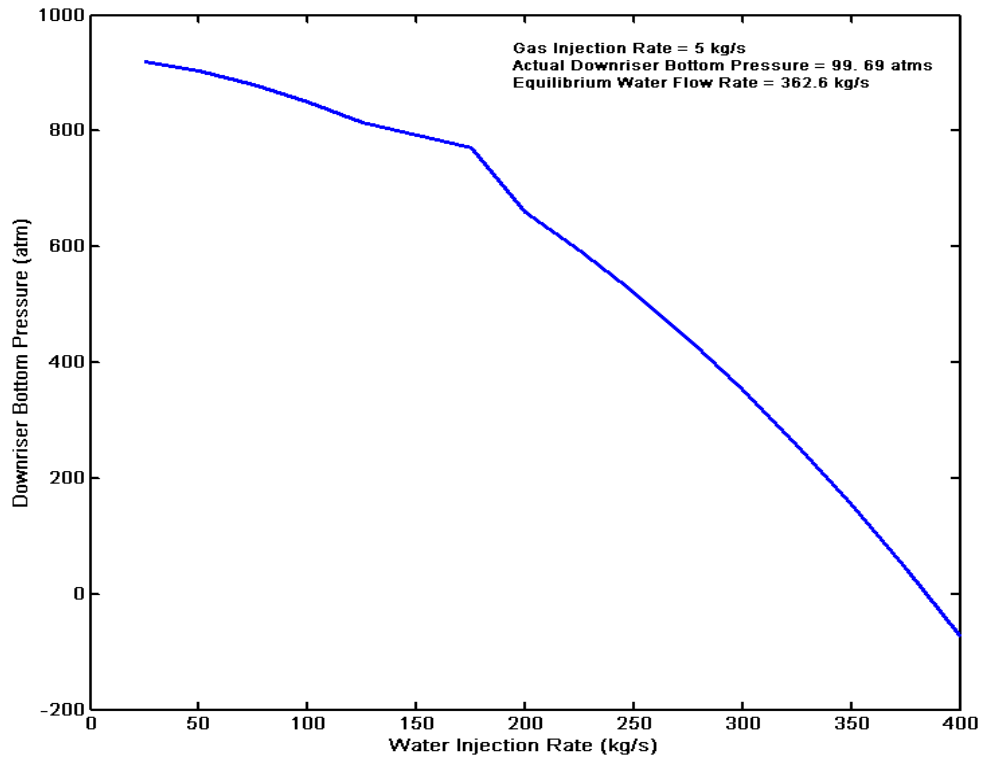


Figure 5-30: P-GLAD water injection rate and corresponding downriser bottom pressure (0.5 m diameter pipe, 200 m upriser, 900 m downriser, injection depth at 300 m)

The P-GLAD system modeled in Figure 5-30 consists of a 0.5 m diameter, 200 m long upriser with an injection inlet at a depth of 300 m and a 0.5 m diameter, 900 m long downriser that disposes the fluid at 1000 m. The system transports 362.3 kg/s of water for a gas injection rate of 5 kg/s. Table 5-1 shows the equivalent volumetric flow rates and velocities at both injection and the top of the upriser .

Table 5-1: Predicted Fluid Volumetric Flow Rates and Velocities for the P-GLAD System

	$Q_g \text{ (m}^3\text{/s)}$	$V_{sg} \text{ (m/s)}$	$Q_L \text{ (m}^3\text{/s)}$	$V_{sL} \text{ (m/s)}$
Injection (300m)	0.066	0.339531	0.294188	1.498
Top of Upriser (100 m)	0.3595	1.802	0.352545	1.7954

5.2.2 Predicted Dissolution Capability

Saito *et al.* (2001) report that the laboratory scale P-GLAD system dissolves over 95% of injected CO₂. These results are for the injection of 95% CO₂ gas. A later study suggests that the length of dissolution pipe necessary to dissolve 99.5% or more of the injected CO₂ increases with an increase in gas impurity levels (Saito *et al.*, 2001). For a 50 kg/s CO₂ injection rate and an 15 m dissolution pipe the following correlation is used to determine the dissolution pipe length:

$$L_p = (58.7(1 - \phi) + 4.7)D_b \times 10^3 \quad [m] \quad (5-1)$$

where ϕ CO₂ gas ratio. Figure 5-31 plots the dissolution pipe lengths predicted by equation 5-1 to the required injection depths for 99.5% CO₂ dissolution in the P-GLAD system.

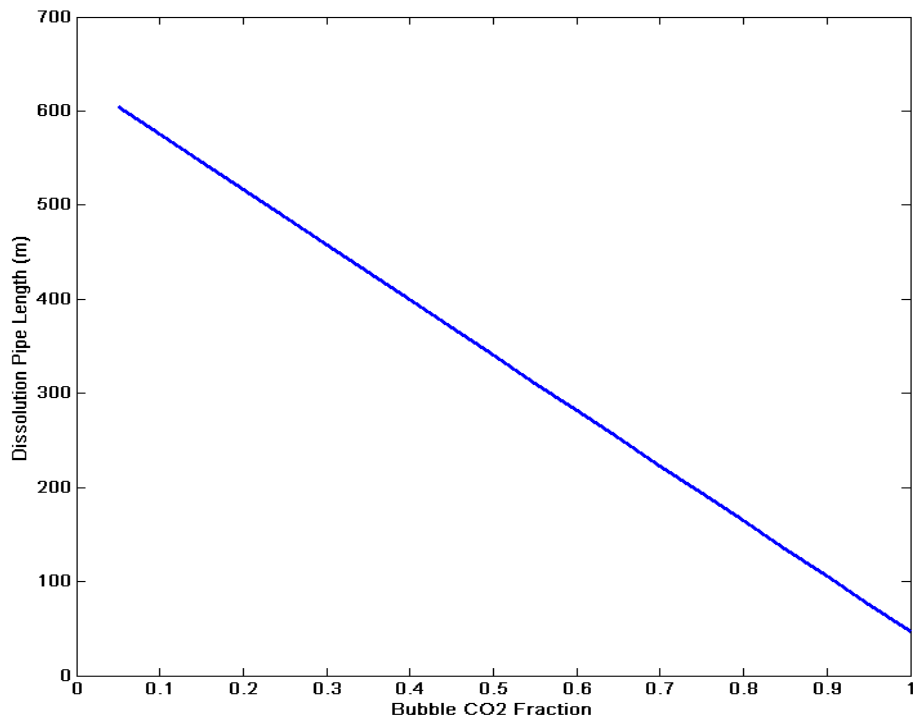


Figure 5-31: Required P-GLAD dissolution pipe length (Saito *et al.*, 2001)

In comparison to Figure 5-31, the single bubble model predicts that even the 15% CO₂ bubble is 99.5% dissolved by the time it reaches the upper mixed level when injected at a depth of 300m. This predicted 200 m dissolution length is much less than that predicted by Saito *et al.* (2001).

5.3 Economic Comparison and Analysis

5.3.1 Cost Factors

Predicting the cost of mixed gas sequestration is difficult due to a lack of available information regarding such processes. The aim of mixed gas injection is to reduce costs by eliminating the necessity of CO₂ separation. Additionally, the hydrate formation zone is decreased by impurities resulting in a range of depths over which lower purity bubbles will exhibit faster dissolution than pure CO₂ bubbles. However, this range of depths is not directly equal to the difference in the hydrate zones, as the mass transfer coefficient is expected to decrease with decreasing concentration of CO₂ in the bubble.

The benefits of injecting low purity CO₂ must be weighed against the costs. Increased impurity and injection depth lead to increased compression costs. Additionally, higher levels of impurities mean higher volumes of gas are necessary to transport a given quantity of CO₂. The result is an increase in transportation costs.

5.3.2. Cost Estimation

Table 5-1 compares the cost of P-GLAD to the cost of direct release of liquid CO₂ as reported by Saito *et al.* (2001). The P-GLAD costs correspond to an injection gas of 95% purity, injected at a depth of 300 m. The direct release costs correspond to the direct release of liquid CO₂ at an unspecified injection depth.

Table 5-2: P-GLAD and Direct Release of Liquid CO₂ Cost Comparison (Saito *et al.*,

	P-GLAD (\$/ton CO₂)	Direct release (\$/ton CO₂)
Separation and capture	26	43*
Liquefaction	0	64
Transportation	30*	7
Sequestration	12	12
Total	68	126

2001)

*Includes compression costs

The compression costs of the direct release method are included in the separation and capture costs. For the P-GLAD process these costs are including in the transportation estimated. The DOE estimates compression costs to be 9\$ per a ton of CO₂, up to pressures of 100 bars (depths of 1000 m) (Wolsky *et al.*, 1994). This yields a compression cost of 9×10^{-3} \$ per a meter of injection depth.

For the direct release of gas mixtures the separation and liquefaction costs are completely eliminated. Furthermore, no change in sequestration costs would arise. The cost of compression can be estimated assuming a linear relationship between the cost and the ratio of the specific volume of the mixture to specific volume of pure CO₂. As the ratio of the specific volume of the mixture to the specific volume of CO₂ increases it becomes more difficult to compress. The density of a mixture is the inverse of its specific volume. Thus the cost of compression can be estimated by

$$Cost = 9.0 \times 10^{-3} \frac{\rho_{CO_2}}{\rho_g} \frac{\$}{m} \quad (5-2).$$

The cost of transportation can be estimated similarly. Assuming the estimated costs of 7\$ per a ton CO₂, corresponds to an injection depth of 1000m, transportation costs for pure CO₂ are

7×10^{-3} \$ per a meter of injection depth. An increase in specific volume of the mixture signifies an increase in the volume of gas that must be transported to dispose of a given quantity of CO₂. Therefore, it is again reasonable to assume that transportation costs for mixtures would be linearly related to ratio of mixture specific volume to pure CO₂ specific volume.

Table 5-3 compares the estimated cost of direct release of 15% CO₂ at 300m to the costs of both the P-GLAD system and the direct release of CO₂. The increase in estimated cost of compression and transportation that result from the increase in impurity exceed the cost decrease in separation. Thus, the direct release of 15% CO₂ has no economic advantage over the P-GLAD process. However, it is cheaper, due to the lack of liquefaction costs, to release 15% CO₂ than to release liquid CO₂. This analysis looks at 15% CO₂, because it is most representative of a flue gas mixture before separation. Further research into separation is necessary to examine the costs of direct release of an intermediate mixture between that of the P-GLAD system (95% CO₂) and flue gas.

Table 5-3: Cost Comparison Including Low Purity Mixture Release

	P-GLAD (\$/ton CO₂)	Pure CO₂ direct release (\$/ton CO₂)	15% CO₂ direct release (\$/ton CO₂)**
Separation and capture	26	43*	60*
liquefaction	0	64	0
transportation	30*	7	46
Sequestration	12	12	12
Total	68	126	118

*Includes compression costs

**1 cm bubbles injected at 1000 m

Chapter 6

6. Conclusion

The work examines the possibility of direct ocean release of CO_2/N_2 mixtures for CO_2 sequestration. It develops a numeric model for the direct ocean release of CO_2/N_2 mixtures for CO_2 sequestration built on past models and experiments of direct release of pure CO_2 . The numeric model is used to simulate ocean release of varying composition mixtures at different injection depths. The simulation results are used to compare gas mixture release to the P-GLAD system for low purity CO_2 sequestration (Saito *et al.*, 2000) and the direct release of pure CO_2 .

The simulation results show that at depths below the hydrate zone low purity mixtures exhibit faster dissolution rates due to the elimination of hydrate formation. At depths above the hydrate formation zone, the initial bubble concentration has little effect on bubble dissolution measured by percent of CO_2 dissolved. It was found that all bubbles injected above 500 m dissolve 95% or more of the injected CO_2 before reaching the upper mixed zone. It is important to note, however, that the absolute mass of CO_2 dissolved decreases with increasing impurity.

It was also shown that low purity bubbles accelerate as they rise due to the expansion of the insoluble gas and the decrease in bubble density. Pure CO_2 bubbles, on the other, decelerate as their bubble diameter shrinks.

The comparison of reported mass transfer coefficients shows the discrepancy in predictions that result from uncertainties in those coefficients. This suggests a need for more experimental work and the development of more accurate mass transfer coefficient correlations, both for pure and mixed gas bubbles.

Finally, this report makes an estimation of the costs of the direct release of a 15% CO₂ bubble injected at 1000 m of 118\$ per a ton-CO₂ injected. This compares to estimated costs of 68 \$/ton-CO₂ for the P-GLAD and 126\$ per a ton-CO₂ for the direct release of liquid CO₂.

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Appendix A

A. MATLAB Code

```
%This code simulates bubble rise and dissolution of a carbon dioxide and nitrogen gas  
%mixture bubble. It requires a user specified bubble diameter, concentration and  
%injection depth It utilizes the SEAWATER algorithms to calculate seawater properties  
%and the Peng Robinson equation of state to determine gas properties.
```

```
%Author: Eric Clower
```

```
%Stanford University
```

```
%August 2002
```

```
clear all variable
```

```
clear
```

```
%Define Constants
```

```
R=83.1451;
```

```
Tc=[304.2 126.2];
```

```
Pc=[72.8 33.5];
```

```
deltat = 2;
```

```
%Initial Conditions
```

```
lat = 36.75;
```

```
D = input('What is initial injection depth?(m)');
```

```
CO2Z = input('What is initial concentration of CO2?');
```

```
N2Z = 1-CO2Z;
```

```
bd = input('What is initial bubble diameter?(cm)');
```

```
bd = bd/100;
```

U0 = 0;

%The main program runs until the concentration of CO2 in the bubble has fallen below a
%control concentration or the bubble reaches the ocean surface

z=[CO2Z N2Z];

n=1;

depthnpo = D;

bdnpo = bd;

CO2MassOrig = 0;

CO2MassNew = 0;

while (depthnpo>0 & bdnpo>0)

time(n) = n * deltat - deltat;

test = 10;

%Sets bubble diameter, depth, rise velocity, and concentration and current time,

%to previously calculated n plus one value

if(n>1)

bd(n) = bdnpo;

Depth(n) = depthnpo;

CO2Z(n) = znpo(1);

U(n) = Unpo;

z = znpo;

Unpo = U(n);

end

if(n==1)

bd(n) = bd;

Depth(n) = D;

CO2Z(n) = CO2Z;

U(n) = U0;

Unpo = U(n) + 0.05;

```

end

z2=[1 0];

%First Determine seawater conditions at current depth
P = Find_P(Depth(n),lat);
T = Find_T(Depth(n));
rowsea(n) = Find_Rowsea(P,T);

%Determine gas mixture conditions at current depth and composition using Peng-
%Robinson equation of state
[x,y,L(n),LiqVol,VapVol,Zv(n),Zl(n)]=flashb(3,z,T,P);
rowbub(n) = Find_Rowbub(x,y,L(n),Zv(n),Zl(n),T,P);
xgs = Calc_Sol(P);

%Numerically solve rise velocity equation
psi = rowsea(n)/(rowbub(n) + 0.5*rowsea(n));
gamma = rowbub(n)/rowsea(n);
vissea = Find_Vissea(T);

%This loop iteratively solves for the new velocity. It starts with an initial guess
%of unpo then it calculates the Reynolds number and solves the equation using
%that Re. It then checks the two Reynolds numbers together
k=1;
while(test > 1)
    %Guess New Velocity
    UnpoGuess = Unpo;

    %Determine Reynolds number and drag coefficient using guessed
    %velocity

```

```

ReGuess = (UnpoGuess*bd(n)*rowsea(n))/(vissea/1000);
Cd = (24/ReGuess)*(1 + 0.173*ReGuess^0.657) + 0.413/(1 +
16300*ReGuess^-1.09);

```

```

%Solve for velocity equation using guessed Drag Coefficient
a = deltat * ((3/4)*psi*(Cd/bd(n)));
b = 1;
c = -U(n) - deltat*((psi*(1-gamma)*9.81));

```

```

p = [a b c];
r = roots(p);
len = length(r);
for(i=1:len)
    if(r(i)>0)
        Unpo = r(i);
    end
end

```

```

%Find Reynold's Number using calculated velocity
ReCalc(n) = (Unpo*bd(n)*rowsea(n))/(vissea/1000);

```

```

%Check if Guessed and Calculated Yield Same Reynold #
test = ReGuess - ReCalc(n);
k=k+1

```

```

end

```

```

%Determine new depth;
depthnpo = Depth(n) - Unpo*deltat;

```

```

%Determine Mass Transfer Out of Bubble
[znpo,massnpo,CO2MassOrig,CO2MassNew(n),MolsLost(n)] =
Calc_CO2Zb(rowbub(n),z,CO2Z(n),bd(n),deltat,rowsea(n),xgs,n,CO2MassOrig,D
epth(n),T,vissea,ReCalc(n),Unpo);
MassLost(n) = (CO2MassOrig-CO2MassNew(n));

%Determine New Bubble Properties and Diameter
bdnpo = Find_NewBubDiameter(depthnpo,lat,znpo,massnpo);

%Find Carbon Dioxide Density
%First perform flash to determine density of CO2 at current depth conditions
[x2,y2,L2(n),LiqVol2,VapVol2,Zv2,Zl2]=flashb(3,z2,T,P);
%Next calculate density from the flash results
rowCO2(n) = Find_Rowbub(x2,y2,L2(n),Zv2,Zl2,T,P);

if(n==1)
    PerCO2Lost(n) = 0;
else
    PerCO2Lost(n) = ((MassLost(n))/CO2MassOrig) * 100;
end

%Increment time step
n = n+1;
end

output = [time' Depth' bd' rowsea' rowbub' rowCO2' CO2Z' MassLost' PerCO2Lost'];
save output.dat output -ascii;

subplot(3,2,1)
plot(time,Depth)

```

```

xlabel('Time(sec)')
ylabel('Depth(m)')
title('Depth of Bubble with Time')
subplot(3,2,3)
plot(Depth,bd)
xlabel('Depth(m)')
ylabel('Bubble Diameter(m)')
title('Bubble Diameter with Depth')
subplot(3,2,2)
plot(Depth,CO2Z)
xlabel('Depth(m)')
ylabel('Fraction CO2')
title('Bubble Carbon Dioxide Content')
subplot(3,2,4)
plot(Depth, rowbub, Depth, rowsea, Depth, rowCO2)
legend('rowbub','rowsea','rowCO2')
title('Densities')
xlabel('Depth(m)')
ylabel('Density (kg/cu m)')
subplot(3,2,5)
plot(Depth,PerCO2Lost)
xlabel('Depth(m)')
ylabel('Percent')
title('Percent CO2 Dissolved');
subplot(3,2,6)
plot(Depth,MassLost)
title('Mass of CO2 Dissolved');
xlabel('Depth(m)')
ylabel('Mass of CO2 Dissolved (kg)');

```

